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Guidelines for the use and interpretation of adsorption isotherm models: A review

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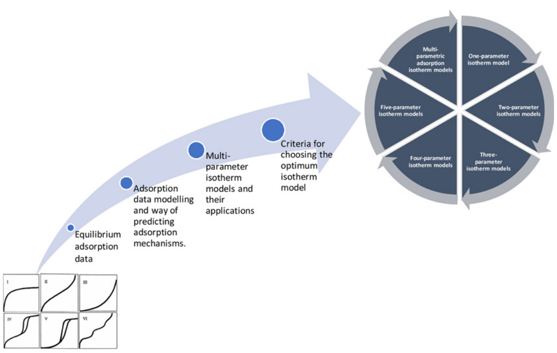
GRAPHICAL ABSTRACT



Abbreviations: a_K, Exponent of Khan model; a_R, Redlich-Peterson isotherm constant (1/mg); a_S, Sips isotherm constant (L/mg); a_T, Toth isotherm constant (L/mg); A_{surf}, Specific surface area of the relevant solid; b, Langmuir isotherm constant or affinity constant (dm^3/mg); b_K, Khan isotherm constant; b_T, Temkin isotherm constant which is related to sorption heat (J/mol); β_S, Sips model exponent; βFS, Fritz0Schlunder isotherm model parameter; b_o, Baudu isotherm equilibrium constant; C_e, Equilibrium concentration of adsorbate on the adsorbent (mg/L); C_{BET}, Isotherm constant which explains the interaction energy with the surface (L/mg); C_{ioc}, Sorbate concentration associated with the natural organic matter (mol.kg^{-1} oc), oc = organic carbon; Ci_{min}, Sorbate concentration associated with the mineral surface (mol.m^{-2}); Ci_{ex}, Ionized sorbate concentration drawn towards positions of opposite charge on the solid surface (mol.mol^{-1} surface charge); Ci_{rxn}, Sorbate concentration bonded in a reversable reaction to the solid (mol.mol^{-1} reaction sites); $Ci_{w,\ neut}$, Concentration of the uncharged chemical in solution (mol/L); $Ci_{w,\ ion}$, Concentration of the charged chemical in solution (mol/L); $C_{s\text{-}BET}$, Isotherm constant which explains monolayer saturation concentration of the adsorbate (mg/L); C, Concentration in (mg/L); F_{oc}, Weight fraction of solid which is natural organic matter (kg oc. kg^{-1} solid); g, Redlich-Peterson exponent that lies between 0 and 1; K_D, Hill isotherm constant; K_F, Freundlich constant; K_{HE}, Henry's isotherm model adsorption constant; K_H, Halsey constant; K_L, Langmuir adsorption constant (L/mg); K_{LF}, Heterogeneous solid equilibrium constant of Langmuir-Freundlich model; K_{FS}, Fritz-Schlunder equilibrium constant (mg/g); K_R, Redlich-Peterson isotherm constant (L/g); K_S, Sips isotherm model constant (L/g); K, Adsorption parameter strength; K_1, K_2, Fritz-Schlunder isotherm parameters; $K_{D\text{-}R}$, Adsorption energy constant in (mol^2/kJ^2); M_{LF}, Heterogeneity parameter which ranges between 0 and 1; M_{MJ}, Describes the spreading of distribution in the path of higher adsorption energies; n_H, Hill cooperativity coefficient of the binding interaction; n_{MJ}, Describes the spreading in the path of lesser adsorption energies; $Q_{\circ}$, Maximum monolayer coverage capacity (mg/g); q_e, Amount of the adsorbate at equilibrium (mg/g); q, Adsorption capacity (mg/g); $q_{m\text{-}BET}$, Isotherm constant which explains theoretical isotherm saturation capacity (mg/g); q_m, Capacity at monolayer coverage (mg/g); q_{MLF}, Langmuir-Freundlich maximum adsorption capacity (mg/g); q_{mFS}, Fritz-Schlunder maximum adsorption capacity (mg/g); q_s, Theoretical isotherm saturation capacity (mg/g); q_{SH}, Hill isotherm maximum uptake saturation (mg/L); N_{AV}, Avogadro's number = 6.022×10^{23}; ν_m, CO_2 adsorption capacity at 273K; R, Universal gas constant (8.314J/mol K); T, Temperature (K); X, Baudu isotherm parameters; E, Dubinin–Radushkevich isotherm constant; α_{FS}, Fritz-Schlunder isotherm model parameter; $\sigma_{\circ}$, Cross-sectional area of CO_2 at 273 which is 21.8 Å; $\sigma_{surf\ ex}$, Net concentration of suitably charged sites on the solid surface (mol surface charge.m^{-2}) for ion exchange; $\sigma_{surf\ rxn}$, Concentration of reactive sites on the solid surface (mol reaction sites.m^{-2}); MB, Methylene blue; MG, Malachite green; CR, Congo red; AR 18, Acid Red 18; MO, Methyl orange; OP, Orange peel; AC, Activated carbon; CAC, Commercial activated carbon; RODP, Roasted date pits; OMSR, Olive mill solid residues; HNH, Hazelnut husk; SCB, Sugarcane bagasse; PHC, Peanut husk charcoal; FA, Fly ash; NZ, Natural zeolite; ATS, Activated tamarind seeds; ANL, Activated neem leaves; SD, Sawdust; LBL, Lignin-based black liquor; FE, Fuller's earth; PFB, Palm-fruit bunch

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ABSTRACT

Adsorption process is considered as one of the most used separation and purification processes, in which adsorption occurs by the formation of the physical or chemical bonds between a porous solid medium and a mixture of liquid or gas multi-component fluid. By taking into consideration the equilibrium data and the adsorption properties of both the adsorbent and the adsorbate, adsorption isotherm models can describe the interaction mechanisms between the adsorbent and the adsorbate at constant temperature. Therefore, understanding modelling of the equilibrium data is a very essential way of predicting the adsorption mechanisms of various adsorption systems. Furthermore, adsorption isotherms in batch experiments can be used for the determination of the solid-water distribution coefficient (K_{id}). This review paper discusses the guidelines of using mono/multi-parametric isotherm models with different applications. The aim of this paper is to establish criteria for choosing the optimum isotherm model through a critical review of different adsorption models and the use of various mathematically error functions such as linear regression analysis, nonlinear regression analysis, and error functions for adsorption data optimization. In this paper, 15 mono-parametric adsorption isotherm models having one, two, three, four and five parameters were investigated. In addition, 10 multi-parameter isotherm models were reviewed as well as addressing their applications.

1. Introduction

For a very long time, the adsorption phenomenon has been known to human beings. It is increasingly used for purification or separation purposes. Adsorption process as a surface phenomenon usually has a porous solid medium as the heart of the process in which a mixture of multi-component fluid (liquid or gas) is attracted to the solid surface either by chemical or physical bonds. The solid porous medium provides high micropore volume; leading to high adsorptive capacity. The porous solid medium has pores that are very small in size wherein adsorbate molecules should find their way to the micropore volume. Since the adsorption process depends on the adsorbent, it should have a good adsorption capacity and kinetics. Adsorbents with small pore size, reasonable porosity, high micropore volume, as well as large pores' network, allowing the molecules to reach the interior of the adsorbent, make the adsorbent suitable for the successful adsorption process (Kong and Adidharma, 2019; Ayawei et al., 2017; Mahmoud et al., 2012; Schwarzenbach et al., 2003a,b; Do, 1998). Moreover, the most important information for a better understanding of the adsorption process is the adsorption equilibria information. The adsorption equilibria of pure components, regardless of the number of components present in the system, is very crucial to understand how much of these components can fit on the solid adsorbent (Ayawei et al., 2017). This piece of information is utilized in studying the adsorption kinetics of single and multicomponent systems, as well as studying the adsorption equilibria of multicomponent systems.

Adsorption isotherm describes the equilibrium performance of adsorbents when the temperature is constant. It depends on the adsorbed species, adsorbate, adsorbent, and various physical properties of the solution including pH, ionic strength, and temperature (Yan et al., 2017). Generally, adsorption isotherms are established when a contact occurs between adsorbate and the adsorbent for sufficient time, in which the interface concentration should be in dynamic balance with the adsorbate concentration existing in the bulk solution. Adsorption isotherms are usually used for designing the industrial adsorption process as well as characterization of the porous solids (Keller and Staudt, 2005). The most important information to understand the adsorption process properly is the adsorption equilibrium information. The great significance of adsorption isotherms of the porous materials is due to its ability in carrying crucial equilibrium information which is needed in various industrial fields like gas storage, CO_2 capture, chemical separation, catalysis, etc. (Kong and Adidharma, 2019; Azizian et al., 2018). Gas adsorption principle depends on the statistical mechanical and theoretical kinetics concepts. According to Lowell et al. (2004), from the various available methods for pore size and surface characterization such as electron microscopy, NMR methods, thermoporometry and others, gas adsorption is the most popular. It has a huge importance in the measurement and characterization of pore structure over a wide range of porous materials ranging from 0.35 nm and > 100 nm including micropores, mesopores and macropores (Flores, 2014; Liu et al., 2019). Demonstrating a curve between how much adsorbate is adsorbed onto the adsorbent and the pressure (in case of gas) or the concentration (in case of liquid) at constant temperature can represent the isotherm models. Generally, prediction of the overall adsorption behavior can be done by modeling the isotherm data by linear analysis as an alternative mathematical approach. However, it was found that the discrepancy between the experimental data and predictions could be caused by linear analysis of the isotherm data. Meanwhile, researchers concluded that parameter estimation by linear analysis is easier (Chen, 2015). Since there is a variation in the interaction between the adsorbent and adsorbate, there will be various isotherm models; consisting of different model parameters (Awad et al., 2019). The present work is aimed to i) review the mono-parametric adsorption isotherm models having one, two, three, four and five parameters, ii) review of multi-parametric isotherm models and applications of linear and non-linear forms, iii) applications of mono-parametric and multi- parametric isotherm models, and iv) discuss the criteria for choosing the optimum mono-parametric isotherm model and error analysis.

2. Adsorption and adsorption isotherm models

The association of chemicals with solid phase is generally known as sorption. The difference between absorption and adsorption is that in absorption the molecules penetrates a three-dimensional matrix, while in adsorption the molecules attach to a two-dimensional matrix (Qi et al., 2017; Prasad and Srivastava, 2009; Nouri et al., 2007). Adsorption is usually described as chemisorption or physisorption process based on the interaction strength between the adsorbate and the substrate (Sims et al., 2019). Physisorption occurs due to the weak electrostatic interactions including London forces, Dipole-dipole forces, and Van der Waals interactions where the bands can be easily broken because of the weak interactions. While chemisorption happens when covalent bond forms between the adsorbate and the substrate by either sharing or transferring of electrons. The interactions of chemisorption are two orders of magnitude stronger than that for physisorption. In chemisorption, a monolayer of adsorbate on adsorbent forms while in physisorption it takes place with the multilayer formation of adsorbate on adsorbent. Physisorption has low enthalpy and occurs at low temperature below the adsorbate boiling point, and it is reversible.

Chemisorption has high enthalpy and occurs at all temperatures and it is irreversible (Singh and Kaushal, 2017).

Moreover, isosteric heat is considered as one of the basic quantities of the adsorption studies. It represents the ratio of the change in the infinitesimal of the adsorbate enthalpy to the infinitesimal change in the adsorbed amount (Azahar et al., 2018). The importance of the information related to the heat released in the kinetic studies is that adsorption causes the release of heat; leading to partly absorption of the released energy by the solid adsorbent as well as partly dissipated to the surrounding. The particle temperature increases by the adsorbed portion of the solid adsorbent leading to slowing down the adsorption kinetics. This can be attributed to the uptake of mass that is controlled by the cooling rate of the particles present in the adsorption later course (Do, 1998). Furthermore, the isosteric heat is calculated from the thermodynamic van't Hoff equation,

$$\Delta H^{o} / R_{g}T_{2} = -(\delta \ln P / \partial T)_{C\mu} \quad (1)$$

Where, ΔH^{o} is the differential molar enthalpy of the adsorption in (J/mol), C_{μ} is the maximum adsorbed concentration, δ is the coefficient of the thermal expansion of the saturation concentration. If C_{μ} decreases with temperature, the isosteric heat will increase. The parameter δ must be zero, so the isosteric heat could take a finite value at high coverage. Here, the saturation capacity does not depend on the temperature, and this leads to a constant heat of adsorption. Exploring novel adsorbents to obtain an ideal system of adsorption requires the establishment of an appropriate adsorption equilibrium correlation, which is considered as an absolute necessity to predict the parameters of adsorption as well as to quantitatively compare the adsorbent behavior for various experimental conditions (Mahmoud et al., 2012).

Additionally, in natural environments, due to the presence of mixture of solids, the sorbate-sorbent association is governed by a combination of interactions. For example, as shown in Fig. 1, 3,4-dimethylaniline compound form some cations of 3,4-dimethyl ammonium after reacting in an aqueous solution. Additionally, the remaining uncharged molecules might penetrate the natural organic matter that is present in the system to escape the water system. Water molecules near the mineral surface might be displaced by the nonionic molecule and be held by polar interactions and London dispersive. Mutual reactive moieties are exhibited by the sorbate and the sorbent due to the ionizable state of the sorbent in the aqueous medium. This will be promoting sorption of the ionic species and leading to the bonding between portions of a chemical to the solid (Al-Asheh et al., 2000; Schwarzenbach et al., 2003a,b).

In addition, adsorption isotherm models by taking into consideration both the equilibrium data and the adsorption properties describes the interaction mechanisms of pollutants and adsorbent materials. According to the classification of IUPAC (International Union of Pure and Applied Chemistry), adsorption isotherms can be categorized into six types based on the isotherm shape of adsorbate-adsorbate pairs as shown in Table 1 (Keller and Staudt, 2005). As shown in Table 1, hysteresis loops have been observed in different IUPAC classified isotherms indicating the mesoporous material presence where adsorption occurred with capillary condensation (Donohue and Aranovich, 1998; Thommes et al., 2015). According to the independent domain theory by Everett (1967), the relative vapor pressure when filling the pores should be larger than that when the pore empties, and this is considered as the fundamental cause of the hysteresis. According to the network theory, hysteresis can also be increased by the pore-blocking effect (Mason, 1983).

During the last decade, three fundamental approaches formulated a wide variety of isotherm models such as Langmuir, Freundlich, Dubinin-Radushkevich, Temkin, Toth and various more isotherms. First approach considers the kinetics where the adsorption and desorption rates are equal, and the adsorption equilibrium will be defined as a state of dynamic equilibrium. The second approach is based on the thermodynamics, providing a framework deriving various adsorption isotherm models with different forms. Moreover, the main idea of generating the characteristic curve is conveyed by the third approach (Foo and Hameed, 2010).

In the literature, great efforts have been made for modeling the adsorption isotherms. Henry, Langmuir, and Toth isotherm models, which are used in the Henry's region, are not valid at high pressure (von Gemmingen, 2005; Tóth, 1994; Foo and Hameed, 2010). As a result, an isotherm model was proposed by Toth, where used the power function of the relation between adsorption capacity and the adsorption potential of adsorbent surface. The same concept was followed by Jaroniec and Marczewksi model (Marczewksi and Jaroniec, 1983; Jaroniec and Marczewski, 1984a; Jarnoiec and Marczewski, 1984b; Tóth, 1981). It was discovered that Toth model failed in the explanation of the transition to the saturation region. Dubinin-Radushkevich isotherm model can perform well at high pressure and failed under low-pressure experimental data (Astakhov and Dubinin, 1971; Bering et al., 1971). On the other hand, Freundlich, Sips, and Jovanovich isotherm models followed the homotattic patch approximation proposed theory by Ross and Olivier (1964). It divides the surface of heterogeneous adsorbent into small-sized homogenous patches, in which the energy distribution of the adsorption energy sites is known.

2.1. One-parameter isotherm model

2.1.1. Henry's isotherm model

According to Ayawei et al. (2017), Henry's isotherm model is considered as the simplest adsorption isotherm model as the partial pressure of the adsorptive gas is proportional to the amount of surface adsorbate. An appropriate fit of the adsorbate adsorption at low concentration is described by Henry's adsorption isotherm model in which all the adsorbate molecules are secluded from their closest neighbors. Consequently, the equilibrium concentrations of the adsorbate in the liquid phase and the adsorbed phases are described by the following linear expression 1:

$$q_e = K_{HE} C_e \quad (1)$$

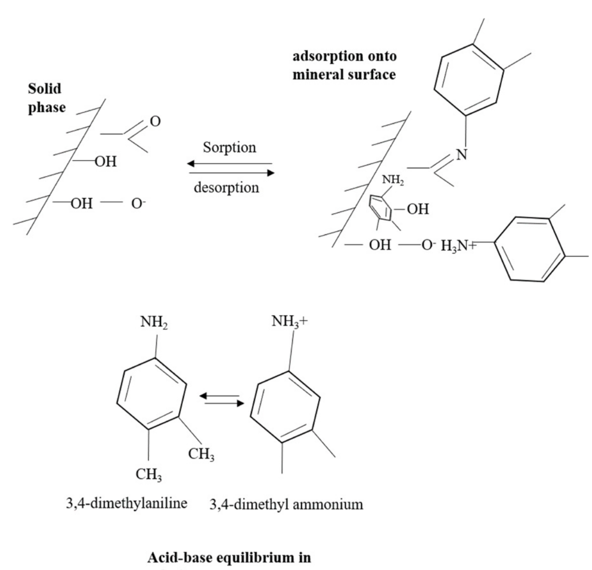
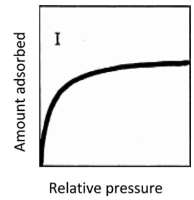
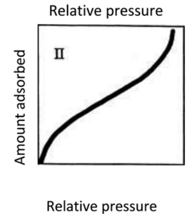
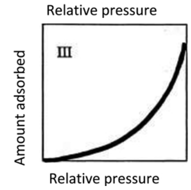
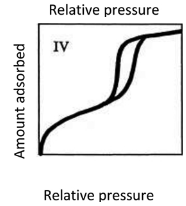
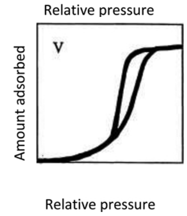
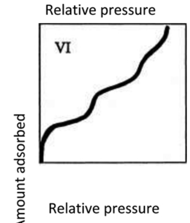


Fig. 1. The association of 3,4-dimethylaniline with natural solids being controlled by some sorbent-sorbate interactions (Modified from: Schwarzenbach et al., 2003a).

Table 1
Categorize of adsorption isotherms.

Isotherm type	Description	Example	Sample graph	References
Type I isotherms (convex upward)	Characterized by a horizontal plateau in which it maintains for very high gas pressures, and it can be described by Langmuir equation.	Adsorption of water vapor on zeolite, adsorption of hydrogen on charcoal. Iodine adsorption from potassium iodine solution on zinc chloride AC		(Inglezakis et al., 2018; Khalfaoui et al., 2003; Keller and Staudt, 2005)
Type II isotherms	Describes adsorption on mesoporous monolayer materials at low pressure and on mesoporous multilayer material at high pressure near saturation with no hysteresis. It has one inflection point. Moreover, its observed in only microporous, nonporous, or disperse solids with > 50 nm pore diameter.	Adsorption of nitrogen on silica gel or iron catalyst, adsorption of water vapor on polymer based adsorbent		(Sultan et al., 2018; Khalfaoui et al., 2003; Keller and Staudt, 2005)
Type III isotherms (concave upward)	This type occurs where the adsorbate–adsorbate interaction is big compared to adsorbate–sorbent interaction.	Adsorption of water on hydrophobic zeolites and activated carbon, adsorption of bromine and iodine on silica gel, carbon tetrachloride adsorption on mesoporous gel		(El Alouani et al., 2019; Khalfaoui et al., 2003; Keller and Staudt, 2005)
Type IV isotherms	This type describes specific mesoporous materials adsorption behavior showing the pore condensation the hysteresis which occurs between the desorption and the adsorption branch. It has 2 inflection points.	Adsorption of humid air, water vapors on specific types of activated carbon, adsorption of benzene on iron oxide and on silica gel.		(wang et al., 2020; Inglezakis et al., 2018; Khalfaoui et al., 2003; Keller and Staudt, 2005)
Type V isotherms	This type indicates the presence of mesopores in phase change like pore condensation could occur. It has one inflection point.	Adsorption of water on carbon molecular sieves and on activated carbon fiber.		(Inglezakis et al., 2018; Khalfaoui et al., 2003; Keller and Staudt, 2005)
Type VI isotherms	At low temperatures, layers will become more pronounced and the isotherms presents stepwise multilayer adsorbates. It has several inflection points.	Adsorption of noble gases on the surfaces of planar graphite, and adsorption of butanol on aluminum silicate, adsorption of CH_4 on MgO		(Sultan et al., 2018; Kong and Adidharma, 2019; Khalfaoui et al., 2003; Keller and Staudt, 2005)

2.2. Two-parameter isotherm models

2.2.1. Langmuir isotherm model

This adsorption isotherm model was originally developed for the description of gas adsorption on solid phase adsorbate like activated carbon (Foo and Hameed, 2010). According to Langmuir theory, adsorption process onto a solid surface is based on a kinetic principle in which a continuous bombardment process of molecules onto the surface with corresponding molecules' desorption or evaporation from the surface with zero accumulation rate at the surface (Langmuir, 1916). In other words, rates of adsorption and desorption should be equal. Traditionally, the Langmuir isotherm model has been utilized for quantifying and contrasting adsorption capacity of various bio-sorbents (Kundu and Gupta, 2006; Foo and Hameed, 2010). Six different and simple adsorption mechanisms were identified and classified by Langmuir due to the surface chemistry diversity and structural geometry of solid materials as shown in Fig. 2 (Swenson and Stadie, 2019). These classifications are (Langmuir, 1918): (i) single-site Langmuir adsorption, the simplest case of gas-solid adsorption, where the surface has identical elementary adsorption sites able to host single adsorbed molecule; (ii) multisite Langmuir adsorption, where more than one type of elementary adsorption sites are available on the surface and each site might fit for single adsorbed molecule. In this case the binding sites are independent and the interactions between adsorbate and adsorbent are neglected; (iii) generalized Langmuir adsorption, where an amorphous material that is treated as a continuum might consist of an intractable number of various adsorption sites with various adsorbate affinities. Since adsorbate-adsorbent interactions are negligible, adsorption isotherm follows the binding energy distribution of the adsorption sites; (iv) cooperative adsorption, in this case binding site on the surface is identical but can host multiple molecules. The energetics of further adsorption is affected by the presence of various adsorbates on the same adsorption site; (v) dissociative adsorption, in this case adsorption is assumed to be 2 fold process: chemical bonding will cause residence at the surface adsorption site and molecular dissociation, then it will undergo desorption where 2 neighboring atoms on the surface have to re-

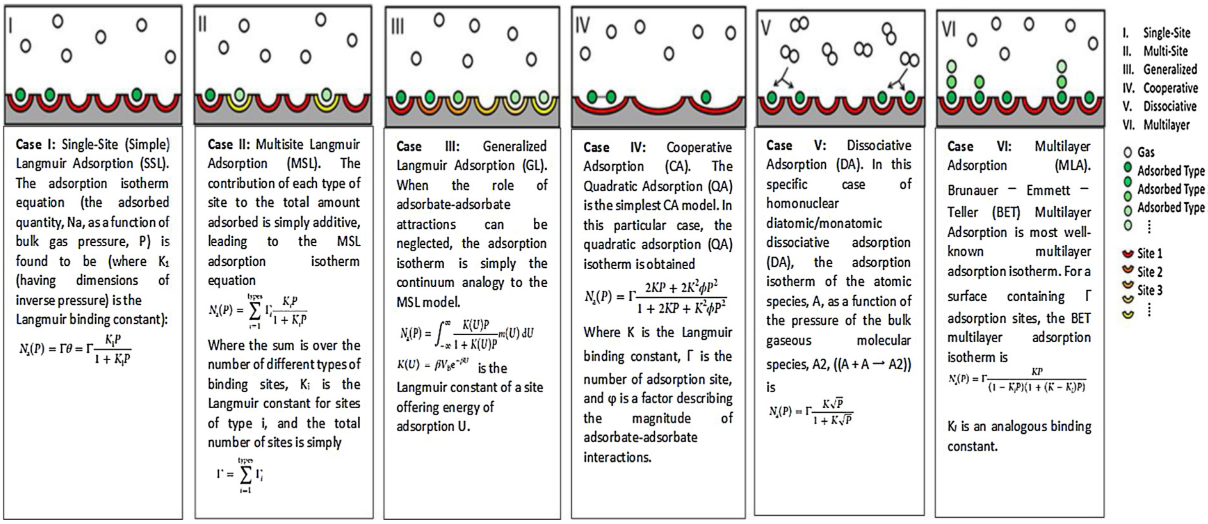


Fig. 2. Schematic representation of Langmuir's six adsorption classifications, modified from (Swenson and Stadie, 2019).

Table 2
Illustration of the linear forms, nonlinear forms and plot of various isotherm models.

Isotherm model	Linear form	Non-linear form	Plot	Reference
One parameter model				
Henry	$q_e = K_{HE}C_e$	–	q_e Vs. C_e	(Ayawei et al., 2017)
Two parameter models				
Langmuir	$\frac{C_e}{q_e} = Q_o - \frac{q_e}{bC_e}$	$q_e = \frac{Q_o bC_e}{1+bC_e}$	$\frac{C_e}{q_e}$ Vs. Q_o	(Al-Ghouti and Razavi, 2020)
	$q_e = Q_o - \frac{q_e}{bC_e}$	–	q_e Vs. $\frac{q_e}{bC_e}$	
	$\frac{1}{q_e} = \frac{1}{Q_o} + \frac{1}{bQ_oC_e}$	–	$\frac{1}{q_e}$ Vs. $\frac{1}{C_e}$	
	$\frac{q_e}{C_e} = bQ_o - bq_e$	–	$\frac{q_e}{C_e}$ Vs. q_e	
Freundlich	$logq_e = logK_F + \frac{1}{n}logC_e$	$q_e = K_F C_e^{\frac{1}{n}}$	$logq_e$ Vs. $logC_e$	(Ayawei et al., 2017; Chen, 2015; Do, 1998)
Dubinin-Radushkevich	$lnq_e = q_s - K\varepsilon^2$	$q_e = q_s e^{-K\varepsilon 2}$	lnq_e Vs. ε^2	(Günay et al., 2007)
Temkin	$q_e = \frac{RT}{bT}\ln A_T + \left(\frac{RT}{bT}\right)\ln C_e$	$q_e = \frac{RT}{bT}\ln A_T C_e$	q_e Vs. $\ln C_e$	(Samarghandi et al., 2009; Vadi et al., 2013)
Flory-Huggins	$log\left(\frac{\theta}{Co}\right) = logK_{FH} + n_{FH}\log(1-\theta)$	$\frac{\theta}{Co} = K_{FH}(1-\theta)^{nFH}$	$log\left(\frac{\theta}{Co}\right)$ Vs. $\log(1-\theta)$	(Almalike, 2017)
Hills	$log\left(\frac{q_e}{q_{SH}-q_e}\right) = n_H logC_e - \log K_D$	$q_e = \left(\frac{q_{SH}\cdot C_e^{nH}}{K_D + C_e^{nH}}\right)$	$log\left(\frac{q_e}{q_{SH}-q_e}\right)$ Vs. $logC_e$	(Koopal et al., 1994)
Halsey	$\ln q_e = \left(\frac{1}{n_H}\right)lnK_H - \left(\frac{1}{n_H}\right)\ln C_e$	$q_e = e^{/nH}_{[lnK_H - lnCe]}$	lnq_e Vs. lnC_e	(Gholitabar and Tahermansouri, 2017; Dada et al., 2017)
Jovanovich	$\ln q_e = lnq_{max} - K_J C_e$	$q_e = q_{max}(1 - e^{K_J C_e})$	lnq_e Vs. C_e	(Saadi et al., 2015)
BET	$\frac{C_e}{q_e(C_s - C_e)} = \frac{1}{q_m C_{BET}} + \frac{(C_{BET}-1)}{q_m C_{BET}}\frac{C_e}{C_s}$	$q_e = \frac{q_e C_{BET} C_e}{(C_s - C_e)\left[1+(C_{BET}-1)\left(\frac{C_e}{C_s}\right)\right]\left[1+(C_{BET}-1)\left(\frac{C_e}{C_s}\right)\right]}$	$\frac{C_e}{q_e(C_s - C_e)}$ Vs. $\frac{C_e}{C_s}$	(Foo and Hameed, 2010)
Three parameter models				
Redlich-Peterson	$\ln\left(K_R\frac{C_e}{q_e} - 1\right) = g\ln(C_e) + \ln(a_R)$	$q_e = \frac{K_R C_e}{1 + a_R C_e^g}$	$ln\frac{C_e}{q_e}$ Vs. lnC_e	(Foo and Hameed, 2010)
Toth	$\ln\left(\frac{q_e}{K_T}\right) = \ln(C_e) - \frac{1}{t}\ln(a_T + C_e)$	$q_e = \frac{K_T C_e}{(a_T + C_e)^{\frac{1}{t}}}$	$\ln\left(\frac{q_e}{K_T}\right)$ Vs. $\ln(C_e)$	(Ho et al., 2002)
Sips	$\beta_S\ln(C_e) = -\ln\left(\frac{K_S}{q_e}\right) + \ln(a_S)$	$q_e = \frac{K_S C_e^{\beta}S}{1 + a_S C_e^{\beta}S}$	$\ln\left(\frac{K_S}{q_e}\right)$ Vs. $\ln(C_e)$	(Saadi et al., 2015)
Khan	–	$q_e = \frac{q_S b_K C_e}{(1 + b_K C_e)^{aK}}$	–	(Khan et al., 1997, 2010)
Four parameter models				
Fritz-Schlunder	–	$q_e = \frac{q_{mFS5}K_{FS}C_e}{1 + q_m C_e^{MFS}}$	–	(Hamdaoui and Naffrechoux, 2007; Nebaghe et al., 2016)
Baudu	–	$q_e = \frac{q_m b_0 C_e^{1+x+y}}{1 + b_0 C_e^{1+x}}$	–	(El Nemr et al., 2010; Hamdaoui and Naffrechoux, 2007)
Weber-Van Vliet	–	$C_e = P_1 q_e^{(P_2 q_e^{P3} + P_4)}$	–	(Ramadoss and Subramaniam, 2018)
Marczewski-Jaroniec	–	$q_e = q_{MMJ}\left(\frac{(K_{MJ}C_e)^{nMJ}}{1 + (K_{MJ}C_e)^{nMJ}}\right)^{MMJ/nMJ}$	–	(Ramadoss and Subramaniam, 2018)
Five parameter models				
Fritz-Schluder	–	$q_e = \frac{1_{mFS}K_1 C_e^{\alpha FS}}{1 + K_2 C_e^{\beta FS}}$	–	(Hamdaoui and Naffrechoux, 2007; Vijayaraghavan, 2015)

associate into a diatomic molecule and leave the surface; (iv) multilayer adsorption, in this case it is assumed that each adsorption site is independent and identical, and molecules are permitted to adsorb above each other as there is no limit for the number of adsorbed molecules.

Langmuir isotherm is an empirical model that assumes that the thickness of the adsorbed layer is one molecule (monolayer adsorption) in which adsorption process occurs at identical and equivalent definite localized sites. There should be no steric hindrance and lateral interaction, even on adjacent sites, between the adsorbed molecules (Vijayarachavan et al., 2006). The Langmuir isotherm model assumes that adsorption is homogenous in which sorption activation energy and constant enthalpies are possessed by each molecule. All sites should have equal affinity towards the adsorbate, as well as no adsorbate transmigration in the surface plane (Kundu and Gupta, 2006). Based on Langmuir theory, there is a relation between the rise of distance and the rapid decrease of the intermolecular attractive forces (Allen et al., 2004; Demirbas et al., 2008). The linear and non-linear mathematical expressions of the Langmuir isotherm model are shown in Table 2.

Furthermore, Weber and Chakravorti (1974), defined the separation factor (R_L) as a dimensionless constant which is represented as:

$$R_L = \frac{1}{1 + K_L C_o} \tag{2}$$

Where, C_o refers to the initial concentration of the adsorbate in (mg/L) and K_L is Langmuir constant which is related to the adsorption capacity in (mg/g). A variation of the suitable area and the adsorbent's porosity can be correlated with K_L constant, implying the fact that higher adsorption capacity can be resulted from large surface area and pore volume. The adsorption nature is indicated by the separation factor to either be linear ($R_L = 1$), irreversible ($R_L = 0$), unfavorable ($R_L > 1$), or favorable ($0 < R_L < 1$).

The Langmuir isotherm equation can be modified for aqueous-phase adsorption as follows (Parker and Garth, 1995):

$$q = q_m \cdot \frac{K_L C}{1 + K_L C} \tag{3}$$

At low concentration and low pressure, the isotherm equation reduces to Henry's law isotherm. A linear increase of the adsorbed amount occurs with pressure, and the saturation capacity of the amount adsorbed is reached when the pressure is sufficiently high leading to the monolayer coverage. Larger affinity constant (b), leads to more surface coverage with the adsorbate molecule due to the strong affinity between them. Since the adsorption process is exothermic process and have positive heat of adsorption, increasing the temperature (T) causes the decrease in the affinity constant. In the adsorption process to occur free energy (ΔG) should decrease and the decrease in the degree of freedom leads to negatively change the entropy (ΔS), thus,

$$\Delta H = \Delta G + T\Delta S < 0 \tag{4}$$

The negativity of the enthalpy (ΔH) indicates that the adsorption process released heat. Similarly, increasing the adsorption heat causes an increase in the adsorbed amount of molecules, which could be attributed to the fact that the adsorbed molecules should overcome a higher energy barrier so they can evaporate back to the gaseous phase. At a given pressure, the adsorbed amount will decrease with increasing the temperature because of the greater energy needed by the adsorbed molecules for evaporation (Jin et al., 2017; Do, 1998).

There are certain flaws in the Langmuir isotherm model in the description of contaminant adsorption onto the sediments due to the fact that sediments are heterogeneous adsorbent with different adsorption energy of each site, and the Langmuir model assumes a homogeneous adsorbent surface having similar adsorption energy for each site (Jin et al., 2017). However, this isotherm model and its coefficients are mostly used by adsorption researchers to determine the contaminants distribution between water and sediments due to the ease of application (Shi et al., 2013; Hafeznezami et al., 2016; Saadi et al., 2013; Jin et al., 2016). The development and application of adsorption technology are limited by the adsorption behavior complexity as well as the completeness and accuracy limitations of the adsorption experimental data. It was concluded by Harter and Baker (1977), that Langmuir equation parameters have the ability to only compare between different adsorbents but fail with the explanation of the reaction mechanism. For the explanation of the sorption surface heterogeneity, site energy distribution (SED) was developed. Its analysis was used to explain the isotherm parameters change and to demonstrate that natural sorbents have a heterogeneous adsorption distribution site (Carter et al., 1995).

2.2.2. Freundlich isotherm model

Freundlich adsorption isotherm model describes the reversible and non-ideal adsorption process. Unlike the Langmuir isotherm model, Freundlich model is not restricted to the monolayer formation in which its application to the multilayer adsorption is possible. In this isotherm model, adsorption heat and affinities does not need to be uniformly distributed on the heterogeneous surface (Adamson and Gast, 1997). Freundlich isotherm model expression defines the heterogeneity of the surface as well as the exponential distribution of the active sites and the active sites energies. In the past, Freundlich isotherm adsorption model was originally developed for animal charcoal adsorption. It demonstrated that at different concentrations of the solution, the ratio of the adsorbate onto certain adsorbent mass to the solute was not constant. In this context, the summation of the adsorption on each site is the adsorbed amount. First, the stronger binding sites will be occupied and then an exponential decline in the adsorption energy will occur upon completing the adsorption process (Zeldowitsch, 1934). Nowadays, there is a wide application of Freundlich isotherm model in systems that are heterogeneous, like adsorption of organic compounds or species that highly interactive on molecular sieves or activated carbon. This adsorption isotherm model is applicable for systems having heterogeneous surfaces in the gas phase. It provides a narrow range of pressure due to the improper behavior of this isotherm towards Henry's law at low pressure. When the pressure is sufficiently high, it does not have a finite limit. Thus, this adsorption isotherm model is valid to not a wide range of adsorption data. This isotherm model has linearized (Eq. (5)) and non-linearized (Eq. (6)) forms of its equation and its parameters can be found when log q_e is plotted vs. log C_e yielding a straight line with $1/n$ as the slope and log (K_F) as an intercept.

$$\text{The linearized form is: } log q_e = log K_F + \frac{1}{n} log C_e \tag{5}$$

$$\text{While the non-linearized form is: } q_e = K_F C_e \frac{1}{n} \tag{6}$$

The type of the isotherm is indicated by the value of n, in which both K_F and n parameters are dependent on temperature. $1/n$ is the intensity of the adsorption or surface heterogeneity indicating the energy relative distribution and the adsorbate sites' heterogeneity. When $1/n$ is greater than zero ($0 < 1/n < 1$) the adsorption is favorable, when $1/n$ is greater than 1, the adsorption process is unfavorable, and it is irreversible when $1/n = 1$. The irreversibility of the isotherm can be attributed to the fact that pressure or concentration should extremely decrease to low value prior to the desorption of adsorbate molecules from the surface (Ayawei et al., 2017; Chen, 2015; Do, 1998). The equation of Freundlich isotherm model is an empirical equation. It was originally obtained by assuming heterogeneity of the surface in which the energy of adsorption is distributed, and topography of the surface is patch wise. Patch wise means that one patch includes all sites with the same adsorption energy. From this perspective, adsorption energy is the produced energy because of the adsorbent and adsorbate interaction. Patches do not interact with each other and each patch is independent from the other. As mentioned before that K_F and n are temperature dependent, the isosteric adsorption heat can be determined by using van't Hoff equation (Eq. (1)) resulting in the following equation:

$$\Delta H = \left[ln(\delta A_o) + \frac{R_g ß}{A_o}\right]A_o + A_o ln(C\mu) \quad (7)$$

From Eq. (7) it can be concluded that the isosteric heat of Freundlich isotherm model is a linear function of the adsorbed amount logarithm (Ayawei et al., 2017; Do, 1998).

2.2.3. Dubinin-Radushkevich isotherm model

Generally, Dubinin-Radushkevich adsorption isotherm model is used for expressing the mechanism of adsorption with the distribution of Gaussian energy onto the heterogeneous surfaces (Çelebi et al., 2007). This model is usually used for the description of vapors and gases adsorption on microporous sorbents such as activated carbon and zeolites. This model succeeded in fitting high activities of the solute and it suits the intermediate adsorbate concentrations range, which can be attributed to the unrealistic asymptotic behavior that is exhibited. However, Henry's law in not predicted by Dubinin-Radushkevich isotherm model when pressure is low (Theivarasu and Mylsamy, 2011). Unlike Langmuir and Freundlich isotherm models, Dubinin-Radushkevich model is a semiempirical equation in which the adsorption of this model follows the mechanism of pore filling. The assumption of this model is that a multilayer character, which involves Van der Waal's forces, and it can be applied to the physical adsorption processes (Israel and Eduok, 2004). The application of this isotherm model usually is to distinguish between the chemical and physical metal ions adsorption (Dubinin, 1960). According to Günay et al. (2007), this isotherm model depends on temperature. It is considered as a distinguishing and unique feature; accordingly, plotting potential energy square versus the logarithm of the adsorbed amount, suitable data will form the characteristic curve in which it lies on the same curve. Dubinin-Radushkevich isotherm model has both non-linear and linear form equations,

$$\text{The non} - \text{linear form is: } q_e = q_s e^{-K\varepsilon 2} \quad (8)$$

$$\text{While the linear form is:} ln q_e = q_s - K\varepsilon^2 \quad (9)$$

E_D is a parameter that is used to predict the adsorption type in which, $E_D = \sqrt{1/(2\,K)}$, Furthermore, ε can be calculated by the following equation $\varepsilon = RT \ln (P_s/P)$, where P_s is the saturation vapor pressure in (atm.) and P is the adsorbate equilibrium pressure in (atm.) (Hu and Zhang, 2019). Accurately determining the adsorption potential (ε) is a crucial prerequisite for utilizing Dubinin-Radushkevich isotherm model. It reflects the Gibbs free energy change of and adsorbent after the adsorption of a unit of molar mass of the used adsorbate. An extension of this isotherm model has been done in covering the adsorption in aqueous phase. It is the same as the gas adsorption model by replacing the pressure with adsorbate equilibrium concentration (C_e) and solubility (C_s). It can be written as: $\varepsilon = RT \ln[1 + 1/C_e]$, where R, T, and C_e are the gas constant (8.314 J/mol.K), absolute temperature (K), and adsorbate equilibrium concentration (mg/L), respectively (Hobson, 1969; Yang et al., 2006; Geszke-Moritz and Moritz, 2016).

Van't Hoff's equation (Eq. (1)) can be utilized for calculating Dubinin-Radushkevich isosteric adsorption heat, resulting in the following equation (Do, 1998):

$$-\Delta H = \Delta H_{vap} + ßE_o(ln1/\theta)^{\frac{1}{2}} + [(ßE_o)\delta T/2](ln1/\theta)^{\frac{1}{2}} \quad (10)$$

Where, ΔH_{vap} is the Latin vaporization heat, ß is a constant that is a function of adsorptive, δ characterizes a saturation capacity change with respect to the temperature,

$$-\delta = \left(\frac{1}{C_{\mu s}}\right)\left(\frac{dC_{\mu s}}{dT}\right) \quad (11)$$

Subtracting the isosteric heat from the vaporization heat yields the net adsorption heat as shown in Eq. (12).

$$q_{net} = ßE_o(ln1/\theta)^{\frac{1}{2}} + [(ßE_o)\delta T/2](ln1/\theta)^{-\frac{1}{2}} \quad (12)$$

Immersion enthalpy is the released heat when adsorption is taking place in a bulk liquid adsorbate. To calculate the immersion heat, net adsorption heat is utilized due to the absence of phase change, which is associated with vapor condensation to liquid phase since adsorption is already from the liquid phase, where immersion heat is given by the following equation:

$$\Delta h_i = \int_0^1 q_{net}(\theta)d\theta = -(ßE_o)(1 + \delta T)T\left(\frac{3}{2}\right) \quad (13)$$

The isotherm curve shape transforms to the convex curve after being sigmoidal curve, with the decrease in the adsorption energy (K). It corresponds to three isotherm curve types that are classified by Giles et al. (1974), which are S-shaped, L-shaped, and H-shaped. The three types of isotherm curves are classified based on both the curvature and the initial slope of the isotherm curves, which depends on the adsorbent affinity towards the adsorbate. Preliminary information regarding the mechanism of adsorption can be provided by a crucial characteristic, which is the adsorption isotherm shape (Hu and Zhang, 2019).

Using Dubinin-Radushkevich adsorption isotherm model for the description of solute adsorption onto a solution or solid phase was not same as when it was used in gas adsorption. This is due to ignoring the solvent influence, particularly pH of the solution on the solutes' chemical species, as well as the charge of the surface and the functional groups, which dissociate from the adsorbents in solid or solution adsorption system. Hence, this model may not provide accurate mean free energy for distinguishing the type of adsorption process either physical or chemical adsorption when the adsorption system is a solid/solution (Tran et al., 2017).

Al-Ghouti and Sweleh (2019) investigated the adsorptive removal of mercury from water by adsorbents derived from date pits. Physical and chemical modification were developed such as thermal roasting (RDP), sulfur (SMRDP) and silane (SIMRDP) based modifications. The authors studied the most-stable Hg^{2+} covered on the adsorbents surface based on the E_D from the Dubinin-Radushkevich adsorption model. This approach was used to distinguish between the physical and chemical adsorption of Hg^{2+} ions. This could be examined by calculating the mean free energy, E_D per molecule of adsorbate (for removing an ion from its location in the adsorption space to the infinity). In addition, the stability of the adsorbed Hg^{2+} on the adsorbents system can be estimated by its E_D as,

$$\Delta E_D = [E(Hg^{2+} adsorbed\, on\, RDP - (E(RDP) + E(Hg^{2+}))]$$

Where E (Hg^{2+} adsorbed on RDP), E (RDP) and E (Hg^{2+}) is total energies of the Hg^{2+} adsorbed RDP system, the RDP surface and a single Hg^{2+} atom, respectively.

Once the Hg^{2+} adsorbed on the surface of the adsorbent may undergo an equilibrium of reduction and oxidation; $Hg^{2+} \leftrightarrows Hg^{+} \leftrightarrows Hg$). Consequently, the configuration would be changed according to the state of the Hg and the amount of energy required for adsorption. Two main paths for Hg^{2+} redox equilibrium on the adsorbent surface were suggested and depicted in Fig. 3. The Hg^{2+} redox equilibrium pathway-I goes via two steps: Hg^{2+}(ads), Hg^{+}(ads), and Hg(ads). The pathway-II directly goes via one step: Hg^{2+}(ads) and Hg (ads).

2.2.4. Temkin isotherm model

Originally, the Temkin empirical isotherm model was used for the description of hydrogen adsorption onto platinum electrodes present in an acidic solution, which is considered as a chemisorption system. This isotherm model takes into consideration the interaction between the adsorbent and the adsorbate in which it ignores the extremely large and low concentration values. This model assumes that adsorption heat (ΔH_{ads}) as a function of temperature, of all molecules existing in the layer declines linearly rather than logarithmically due to the surface coverage increase (Aharoni and Ungarish, 1977; Vadi et al., 2013). This adsorption isotherm model is only valid for an intermediate

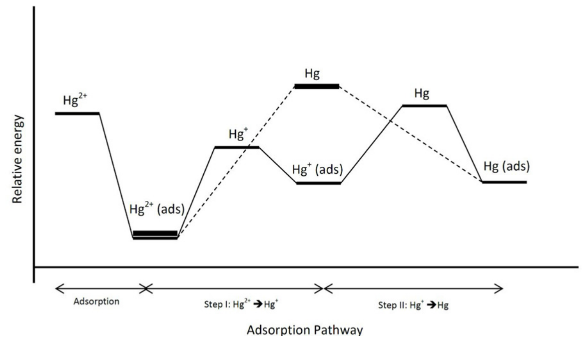


Fig. 3. Adsorption pathways and relative energy profile of Hg^{2+} adsorption on the surface of adsorbent (Al-Ghouti and Sweleh, 2019).

concentration range (Shahbeig et al., 2013). Similarly, to the above-mentioned isotherm models, Temkin model has non-linear and linear forms of its equation, which are:

$$\text{Non-linear form: } q_e = \frac{RT}{b_T} \ln A_T C_e \tag{14}$$

$$\text{While the linear form is: } q_e = \frac{RT}{b_T} \ln A_T + \left(\frac{RT}{b_T}\right) \ln C_e \tag{15}$$

By plotting q_e vs. $\ln(C_e)$, both A_T and b_T constants can be obtained (Samarghandi et al., 2009; Vadi et al., 2013). It is implied in its equation that binding energies are uniformly distributed, and this model is excellent in predicting the equilibrium of the gas phase, in case that it is not necessary for it to be organized in a tightly packed structure with identical orientation. On the other hand, this isotherm model is not appropriate to present complex adsorption systems including aqueous-phase adsorption isotherms.

2.2.5. Flory-Huggins isotherm model

A simple yet powerful mathematical model for the polymer blends' thermodynamics is given by the solution theory of the Flory-Huggins. According to Senichev and Tereshatov (2014), dissolution process is derived by combining both factors of entropy and enthalpy, where it can be described based on Flory-Huggins equation. This theory assumes that lattice sites are occupied by the polymer segments, while solvent molecules occupies single sites. Based on this assumption, the entropy of long-chain mixed molecules can be calculated.

This isotherm adsorption model accounts for the nature of surface coverage degree of the adsorbate on the adsorbent. Flory-Huggins isotherm model describes the nature of the adsorption process regarding the feasibility and spontaneity of the process (Horsfall and Spiff, 2005). The linear and non-linear equations of this adsorption isotherm model are shown in Eqs. ((16) and (17)), respectively. Their parameters can be determined by plotting log (θ/C_o) vs. log ($1 - \theta$), where the yielded straight line confirms the model (Almalike, 2017).

$$\text{Linear form: } log\left(\frac{\theta}{C_o}\right) = logK_{FH} + n_{FH}\log(1-\theta) \tag{16}$$

$$\text{Non-linear form: } \frac{\theta}{C_o} = K_{FH}(1-\theta)^{n_{FH}} \tag{17}$$

In this respect, the surface coverage degree is denoted as (θ), while n_{FH} is the number of metal ions that occupies the adsorption sites on two membranes. In addition, K_{FH} is the adsorption equilibrium constant. It is usually utilized to calculate the spontaneity free Gibbs energy, as it is related to the following expression (Vijayarachavan et al., 2006).

$$\Delta G^{\circ} = -RT\, lnK_{FH} \tag{18}$$

2.2.6. Hills isotherm model

This adsorption isotherm model explains different species' binding onto homogeneous substrates. The assumption of this isotherm model is that adsorption is a cooperative phenomenon, with the adsorbates having the ability to bind at one site on the adsorbent, which could be influencing other binding sites on the same adsorbent. Originally, the equation of this adsorption isotherm model was derived from the non-ideal competitive adsorption (NICA) isotherm (Koopal et al., 1994). The linear and non-linear forms of this isotherm model are expressed as follows:

$$\text{Linear form: } log\left(\frac{q_e}{q_{SH} - q_e}\right) = n_H logC_e - \log K_D \tag{19}$$

$$\text{Non-linear form: } q_e = \left(\frac{q_{SH} \cdot C_e^{nH}}{K_D + C_e^{nH}}\right) \tag{20}$$

Furthermore, in this model if $n_H = 1$ means that binding is hyperbolic or non-cooperative, if $n_H > 1$, binding has positive cooperativity, while negative cooperativity occurs when $n_H < 1$.

2.2.7. Halsey isotherm model

This adsorption isotherm evaluates the multilayer adsorption system and describes its condensation at a relatively large distance from the surface. Similarly, to Freundlich isotherm model, Halsey model is suitable for the multilayer adsorption as well as the heterogeneous surfaces in which the adsorption heat is non-uniformly distributed (Gholitabar and Tahermansouri, 2017). The linearized form of Halsey equation is given as follows:

$$\ln q_e = \left(\frac{1}{n_H}\right) lnK_H - \left(\frac{1}{n_H}\right) \ln C_e \tag{21}$$

While the nonlinearized form is given as follows (Dada et al., 2017).

$$q_e = e^{[lnK_H - lnC_e)/n_H} \tag{22}$$

Halsey isotherm parameters can be obtained from the plot of $\ln q_e$ versus $\ln C_e$, where K_H is obtained from the slope and n_H is obtained from the intercept.

2.2.8. Jovanovich isotherm model

The assumptions of this model are contained in the Langmuir isotherm model with some possible additions of mechanical contact between both the desorbing and adsorbing molecules. The allowance of the adsorption surface of Jovanovich model is made, the equation of this adsorption isotherm model has less usage in the physical adsorption while it can be applied to both mobile and monolayer localized adsorption with no lateral interactions (Saadi et al., 2015). The equation of this model can reach the limit of saturation when the concentration is high, while at low concentration it reduces to Henry's law. Compared to Langmuir equation, Jovanovich equation has a slower approach toward saturation. The linearized form of Jovanovich equation is given by:

$$\ln q_e = lnq_{max} - K_J C_e \tag{23}$$

While the non-linearized form is given by,

$$q_e = q_{max}(1 - e^{K_J C_e}) \tag{24}$$

Where, q_{max} describes the maximum adsorbate uptake and KJ is Jovanovich constant (L/g) in which they can be obtained by plotting $\ln q_e$ versus C_e (Samarghandi et al., 2009).

2.3. BET (Brunauer, Emmett and Teller) isotherm and modified BET isotherms

Brunauer-Emmett-Teller is a theoretical isotherm equation that is most applicable in the equilibrium of gas-solid systems. Multilayer adsorption systems with relative pressure ranging from 0.05 to 0.30, which corresponds to a monolayer coverage between 0.5–1.50, were derived from the development of BET isotherm. The common application of this model is for the determination of an adsorbent's surface area from nitrogen adsorption data (Foo and Hameed, 2010). The linear BET form is expressed as follows:

$$\frac{C_e}{q_e(C_s - C_e)} = \frac{1}{q_m C_{BET}} + \frac{(C_{BET} - 1)}{q_m C_{BET}}\frac{C_e}{C_s} \tag{25}$$

From the linear plot of $\frac{C_e}{q_e(C_s - C_e)} Vs. \frac{C_e}{C_s}$, C_{BET} and q_m can be calculated from the slope and the intercept, respectively. Since C_{BET} indicates the surface binding energy, having negative value is an indication that the isotherm model is inadequate for the explanation of the adsorption process (Savran et al., 2017). The equation can be simplified as both C_{BET} and $C_{BET}(C_e/C_s)$ are greater than 1:

$$q_e = \frac{q_s}{1 - \left(\frac{C_e}{C_s}\right)} \tag{26}$$

This model is extended to the liquid-solid interface and it is described as follows:

$$q_e = \frac{q_{mBET} C_{BET} C_e}{(C_e - C_s)\left[1 + (C_{BET} - 1)\frac{C_e}{C_s}\right]} \tag{27}$$

BET model is considered as a special form of Langmuir model. It has the same assumptions that are applied in the Langmuir model by adding other simplified assumptions, which are: same adsorption energy are found in the second, third and higher layers. This energy is equal to the fusion heat that is not directly affected by the interactions between the adsorbent and the adsorbate. However, the first layer has different energy from that for the other layers. As the concentration reaches the saturation concentration, the number of layers tends to infinity (Saadi et al., 2015).

Surface area of porous materials such as zeolites and metal-organic frameworks (MOFs) is commonly determined through BET analysis. MOFs are microporous materials that possess surfaces that are far from flat, and they exhibit very large BET surface areas. Walton and Snurr (2007), validated the applicability of the BET analysis to MOFs with micropores ranging between 7 and 20 Å by grand canonical Monte Carlo (GCMC) simulation study to predict nitrogen adsorption isotherms in MOFs series. Another GCMC simulating study validated the applicability of the BET method to determine MOFs and zeolite surface areas, even for materials with heterogeneous pores or ultra-micropores (< 7 Å) (Bae et al., 2010). However, for ultra-micropores close to nitrogen and argon kinetic diameter, measuring the isotherms of nitrogen and argon at 77 K and 87 K is impossible due to the inability of molecules to overcome the activation energy, CO_2 isotherms at 273 K are usually used as an alternative way. Employing BET theory to CO_2 isotherm is achieved by a plot between x and x/q(1-x) where x is P/P_o for CO_2 at 273 K and q is the uptake capacity of CO_2 over MOFs. The plotted graph gives [c-1/v_m] and [1/v_mc] as an intercept and slope, respectively. Estimation of the surface area can be done by using: $A = v_m \sigma_o N_{AV}$. Furthermore, two conditions must be fulfilled for the satisfaction of this criterion of ultra-microporous surface area measurement: (i) q(P-P_o) value should increase with increasing x, and (ii) y-intercept in the linear region of the graph must be > 0 to have a meaningful value of c (Ullah et al., 2019).

Frenkel-Halsey-Hill (FHH) is a modified version of BET mode. It is an adsorption isotherm that was derived from the potential theory, which describes the multilayer adsorption. This model assumes that the adsorption potential variation depends on the distance between the surface layer and the adsorbed molecule layer (Saadi et al., 2015). Unlike the BET assumption, FHH assumes that adsorption in all the adsorbed layers is affected by the surface heterogeneity. The adsorption isotherm equation is written as:

$$q_e = \left[\frac{-1}{A_{FHH}} \ln\left(\frac{C_e}{C_s}\right)\right]^{\frac{-1}{B_{FHH}}} \tag{28}$$

Where, A_{FHH} indicates the long-range van der Waals interactions between the first layer of the adsorbed molecule and the surface, as well as the interaction between the neighboring adsorbate molecules, which have information about the surface adsorption capacity. High A_{FHH} value is an indicative that there is a capability of adsorbing more adsorbate molecules. The interaction between the subsequent layers of the adsorbate and the surface are described by B_{FHH} in which the value of this parameter increases with increasing the homogeneity of the sample. B_{FHH} can reach 2.95 for most homogeneous samples and decrease to 2.69 for less heterogeneous samples and further decrease to 2.55 for most heterogeneous samples. The small value of B_{FHH} or large surface heterogeneity is an indicative that adsorbent's influence on the adsorbed molecules is extended at further distances from the surface (Rudzinski and Everett, 1992).

Moreover, MacMillan-Teller (MET) isotherm model was extended based on the interpretation of the effect of surface tension inclusion on the BET isotherm model in which it can be defined as follows:

$$q_e = q_{SM}\left(\frac{k}{\ln\left(\frac{C_s}{C_e}\right)}\right)^{1/3} \tag{29}$$

Where, k is the isotherm constant. When unity is approached by C_s/C_e, the logarithmic term can be approximated as follows:

$$q_e = q_S\left(\frac{kC_e}{C_s - C_e}\right)^{1/3} \tag{30}$$

Unlike BET model, which is only valid for concentrations lower than 0.4, this empirical isotherm model can fit with experimental data with concentrations more than 0.8 (Saadi et al., 2015).

Another modified version of BET model is Aranovich model, which can be expressed as:

$$q_e = \frac{q_{mAr} C_{Ar}\frac{C_e}{C_s}}{\left(1 - \frac{C_e}{C_s}\right)^{\frac{1}{2}}\left(1 + C_{Ar}\frac{C_e}{C_s}\right)} \tag{31}$$

This equation is similar to the equation of BET model, but it differs in the exponent value of (1-C_e/C_s) in which in Aranovich model it is one-half while in the BET equation it is equal to 1. The BET equation describes a narrow concentration range, while Aranovich model has a broader concentration range. This isotherm model has four hypotheses, namely: i) flat and uniform surface of the adsorbent, ii) lattice model can be applied in the vacancy solution which is phase in contact with the adsorbent, iii) the change in energy accompanies the molecule evaporation is based on the layers' number, and iv) this model considers only the configurational components of the free energy (Saadi et al., 2015).

It is assumed by Harkin-Jura isotherm model that it is possible that the multilayer adsorption on the adsorbents' surface to have a heterogeneous distribution of pores. This isotherm model is presented as follows:

$$\frac{1}{q_e^2} = \left[\frac{B_{HJ}}{A_{HJ}}\right] - \left[\frac{1}{A_{HJ}}\right]\log(C_e) \tag{32}$$

The isotherm constants are represented as B_{HJ} and A_{HJ} which can be obtained from the slope and intercept by plotting $1/q_e^2$ versus $\log(C_e)$, respectively (Liu and Wang, 2013). At higher concentrations, the explanation of the high fit adsorption data can be achieved by the existence of the heterogeneous distribution of the pores. Like BET model, this isotherm model is applied to determine the adsorbent surface area (Gürses et al., 2004; Shanavas et al., 2011).

2.4. Three-parameter isotherm models

2.4.1. Redlich – Peterson isotherm model

It is a hybrid isotherm model that features Freundlich and Langmuir isotherm models with three parameters. Since this model is a combination of both models, therefore, the adsorption mechanism does not follow an ideal monolayer adsorption. Redlich-Peterson is a versatile isotherm model that can be applied in heterogeneous or homogeneous systems. This isotherm model has an exponential function in the denominator and in the numerator. It has a linear dependence on the concentration representing the equilibrium of the adsorption onto a wide range of concentration (Prasad and Srivastava, 2009). Since the numerator is from the Langmuir isotherm model, at infinite dilution, it can approach Henry region. The linear and nonlinear equations of this isotherm model are represented at Eqs. (33) and (34), respectively (Foo and Hameed, 2010).

$$\ln\left(K_R \frac{C_e}{q_e} - 1\right) = g\ln(C_e) + \ln(a_R) \tag{33}$$

$$q_e = \frac{K_R C_e}{1 + a_R C_e^g} \tag{34}$$

Plotting $\ln(C_e/q_e)$ versus $\ln(C_e)$ allows the determination of g and a_r values from the slope and intercept, respectively. This model reduces to Freundlich equation at high liquid-phase concentration of the adsorbate (Ayawei et al., 2017),

$$q_e = \frac{K_R}{a_R} C_e^{1-\beta} \tag{35}$$

Where, $K_R/a_R = K_F$ and $(1\text{-}\beta) = 1/n$ of Freundlich isotherm model. However, it reduces to Langmuir equation when $\beta = 1$, $a_R = b$ (Langmuir adsorption constant), and $K_R = bQ_o$, and it reduces to Henry's equation when $\beta = 0$ and Henry's constant is represented by $1/(1 + b)$ (Hutson and Yang, 1997).

This isotherm model adopts a minimized procedure to solve the equations. It maximizes the coefficient of correlation between the predictions of the theoretical model and the points of the experimental data (Wong et al., 2004). Regarding the limits, at the low concentration limit this model approaches the ideal Langmuir condition in which values of β are close to one, and it is in accordance with a high concentration limit of Freundlich isotherm model in which the exponent β tends to be zero (Foo and Hameed, 2010).

2.4.2. Toth isotherm model

This isotherm model is an empirical modified form of the Langmuir equation; aiming to reduce the error between the value of both the experimental and predicted value. Mostly, this isotherm model is utilized for the description of heterogeneous adsorption system; satisfying the low and high concentration of adsorbate. It is presupposed by the correlation of this isotherm model that an asymmetrical quasi-Gaussian distribution of energy, in which most sites have lower energy of adsorption than the mean value or the peak (Ho et al., 2002). The linear and nonlinear forms of this isotherm model can be expressed as follows, respectively,

$$\ln\left(\frac{q_e}{K_T}\right) = \ln(C_e) - \frac{1}{t}\ln(a_T + C_e) \tag{36}$$

$$q_e = \frac{K_T C_e}{(a_T + C_e)^{\frac{1}{t}}} \tag{37}$$

The equation reduces to Langmuir isotherm equation when t = 1. Thus, (t) is a parameter that indicates the heterogeneity of the adsorption system in which it is considered heterogeneous when (t) does not equal the unity (Podder and Majumder, 2016). Furthermore, increasing the temperature leads to the rapid increase of a_T due to the independent relation between temperature and the parameter t.

According to Parker and Garth (1995), Toth equation reduces to Henry's law at low concentration, while at high concentration, approaching a finite capacity is slower than the Langmuir isotherm model. This isotherm model is usually used for the description of various gas adsorption as well as the adsorption of organic compounds' vapors (Keller and Staudt, 2005). Moreover, the equation of the Toth isotherm model has the ability for describing the data behavior at high and low concentrations, which is considered as an advantage over Sips equation.

According to Do (1998), at zero loading, the slope of this isotherm model has a constant limit, while it starts to reduce at a given loading when it is at a much faster rate than that for Langmuir equation. This can be attributed to the effect of the heterogeneity, which indicated in the parameter t. Physically, it is preferred by the molecule to be adsorbed onto the high-energy sites and then with the progress of the adsorption, molecules are adsorbed onto sites with decreased energy, which in turn leads to a slower increase in the adsorbed amount versus pressure, compared to that of the Langmuir equation.

In various applied adsorption problems, it is needed to estimate the site energy distribution of adsorbents for specific molecules. From the isotherm theoretically representing the experimental equilibrium data, the site energy distribution of an adsorbent can be determined. Unlike homogenous systems where it can be determined straightforward, heterogeneous systems require the generation of corresponding affinity distribution from the solution of the adsorption integral $q_e(p) = \int_{E_{min}}^{E_{max}} q_h(E, p) f(E) dE$, assuming the total adsorption of gas molecules. Since solving this equation is difficult due to the lack of general analytical solution, the condensation approximation method can be used. Site energy distribution can be directly defined by the condensation approximation method through the applied isotherm equation. In which the determined energy distribution is controlled by the pressure range used in the experimental isotherm. Determining the site energy distribution by this method allows the analysis of the site energy distribution changes that are associated with the isotherm parameters' changes. For the isotherms following Toth model, energy site distribution function can be obtained from $f(E^*) = \frac{q_m b \exp\left(-\frac{E^*}{RT}\right) p_s \left(\left(p_s \exp\left(-\frac{E^*}{RT}\right)\right)^t + b\right)^{\frac{-t+1}{t}}}{RT}$, where the binding site energy can be determined directly when t equals unity (Kumar et al., 2011).

2.4.3. Sips isotherm model

A combination of both Langmuir and Freundlich isotherm models resulted in the formation of the Sips isotherm model, which is deduced to predict the heterogeneity of the adsorption systems as well as to circumvent the limitations associated with the increased concentrations of the adsorbate of Freundlich model. This, in turn, leads to the production of an expression that have a finite limit at high concentration. Sips model has the validity in localizing the adsorption without the adsorbate-adsorbate interaction (Saadi et al., 2015). Sips equation is given by the following linear and nonlinear expression, respectively.

$$\beta_S \ln(C_e) = -\ln\left(\frac{K_S}{q_e}\right) + \ln(a_S) \tag{38}$$

$$q_e = \frac{K_S C_e^{\beta} S}{1 + a_S C_e^{\beta} S} \tag{39}$$

The Sips isotherm model does not follow Henry's law due to the reduction to Freundlich model when the adsorbate concentration is low. On the other hand, at high adsorbate concentration, it predicts the monolayer adsorption characteristic of Langmuir model. Operating conditions such as alteration of the concentration, pH, and temperature, governs the parameters of the equation (Pérez-Marín et al., 2007). Similarly, to Freundlich model, same disadvantages are shared with sips model in that neither Freundlich nor Sips model gives the correct

Henry's law limit when pressure is low (Do, 1998).

2.4.4. Khan isotherm model

This isotherm model is a generalized model suggested for the adsorbate adsorption from pure solutions, in which it is representing both Langmuir and Freundlich models. The development of this isotherm model was made for single component and multicomponent adsorption systems. The equation of this model is expressed as follows:

$$q_e = \frac{q_S b_K C_e}{(1 + b_K C_e)^{a_K}} \tag{40}$$

The maximum uptake values are well determined at high correlation coefficients as well as minimum chi-square values and sum square error (ERRSQ) value. When the value of the equilibrium concentration is large, Khan model reduces to Freundlich isotherm. On the other hand, this isotherm model is reduced to the Langmuir isotherm system when a_K equals the unity (Khan et al., 1997, 2010).

2.5. Four-parameters isotherm models

2.5.1. Fritz–Schlunder isotherm model

Fritz-Schlunder is an empirical equation of Langmuir-Freundlich type that a wide range of experimental results can fit in this equation due to the large number of coefficients in the isotherm (Hamdaoui and Naffrechoux, 2007; Nebaghe et al., 2016). The equation of this isotherm model can be expressed as:

$$q_e = \frac{q_{mFS5} K_{FS} C_e}{1 + q_m C_e^{MFS}} \tag{42}$$

Where, M_{FS} (Fritz-Schlunder equilibrium model) becomes to Langmuir model if its = 1, however, the model reduces to Freundlich model when the adsorbate concentrations are high. Nonlinear regression analysis can be used to determine the isotherm parameters (Ayawei et al., 2017).

2.5.2. Baudu isotherm model

It was observed that estimating the Langmuir coefficients (b and q_m) by tangents measurements at various equilibrium concentrations revels that these coefficients are not constants in the broad concentration range, consequently, the Langmuir isotherm was reduced to Bauder isotherm as shown in Eq. (43). This isotherm model is applicable in the range of $(1 + x + y) < 1$ and $(1 + x) < 1$. Baudu isotherm model is reduced to Freundlich isotherm model when surface coverage is low, and the represented by the Eq. (44). The parameters of this isotherm can be determined by nonlinear regression analysis (El Nemr et al., 2010; Hamdaoui and Naffrechoux, 2007).

$$q_e = \frac{q_m b_0 C_e^{1+x+y}}{1 + b_0 C_e^{1+x}} \tag{43}$$

$$q_e = \frac{q_m b_0 C_e^{1+x+y}}{1 + b_0} \tag{44}$$

Table 3
Description of adsorption isotherm models that are used for multi-parametric systems.

Adsorption isotherm model	Description	Reference
Extended Langmuir multi-parametric isotherm (EL)	- It assumes that there is a similarity between all the present active sites on the adsorbent surface, and there is equal competition between the species present in the environment for the same active sites. - With respect to the mono-parametric system, the precision of the obtained values is more when it is closer to the species adsorption capacity values.	(Srivastava et al., 2008)
Non-modified Langmuir (N-ML) model	- Similarly, to the EL model, it assumes the similarity between the active sites and species compete equally for the same active sites. - Since this model uses the mono-parametric adsorption parameters, species adsorbed in multi-parametric system is not represented adequately.	(Srivastava et al., 2006)
Modified Langmuir isotherm (ML)	- It evaluates the ions' unequal competition from the adsorbent's active sites through incorporating the η_L constant parameters in its formula. - η_L values are obtained from the multi-parametric adsorption process in which its characteristic for each species. - In comparison with the competing species, species with small value will have greater affinity with the adsorbent's active sites.	(Reynel-Avila et al., 2016)
Extended selective Langmuir isotherm (ESL)	- It assumes that part of the adsorbent surface will be covered by each ion of the existing ions in the mixture. - The summation of all the adsorption capacities for all the species and multiplying it with the molar fraction of the adsorbed species yields the maximum adsorption capacity of the adsorbent.	(Soetaredjo et al., 2013)
Extended Freundlich (EF) multi- parametric isotherm	- Can be applied in non-ideal systems - Better fits of this model depends on the heterogeneity of the adsorbent surfaces allowing various interaction types with the existing species in the solution.	(Padilla-Ortega et al., 2013)
Sheindrof – rebuhn – sheintuch (SRS) isotherm	- It is an extension of Freundlich isotherm model. - It considers the competition between the existing species in the adsorption system and how they affect the adsorption of each other. - It uses the competition coefficient (a_{ij}) of species *i* by component *j* which evaluates the species competition in the system. - It allows the description of the adsorption inhibition by component *j* for species *i*. - - It assumes that there is an exponential distribution of site adsorption energy of each component in the multi- parametric system	(Vijayaraghavan et al., 2016)
Non-modified Redlich – Peterson (N-MR-P) multi-parametric isotherms	- This extended model assumes that there is an equal competition between adsorbent's active sites and the ions.	(Reynel-Avila et al., 2016)
Modified Redlich – Peterson (MR-P) multi-parametric isotherms	- It uses the multi-parametric competition constant ($\eta_{R,i}$) for the evaluation of unequally competitive adsorption. - Compared to competing species, the smaller the species value, the greater the affinity towards the adsorbent's active sites.	(Reynel-Avila et al., 2016)
Extended sips (ES) model	- This model can only be applicable to systems that present active sites with heterogeneous energies.	(Al-Asheh et al., 2000)
Ideal adsorption solution theory – sips (IAST-S) model	- It is applicable to adsorbents with surfaces having homogenous active sites.	(Al-Asheh et al., 2000)

2.5.3. Weber-Van Vliet isotherm model

An empirical relation with four parameters has been postulated by Weber and Van Vliet to describe the equilibrium data for a wide range of adsorption systems as shown in Eq. (45).

$$C_e = P_1 q_e^{(P_2 q_e^{P3} + P_4)} \tag{45}$$

Where, P_1, P_2, P_3, and P_4 are isotherm parameters. The isotherm parameters can be determined by multiple nonlinear curve fitting technique that is predicted on the minimization of sum of square of the residual (Ramadoss and Subramaniam, 2018)

2.5.4. Marczewski-Jaroniec isotherm model

This isotherm model is also known as four parameter general Langmuir equation. The isotherm equation is expressed as follows:

$$q_e = q_{MMJ}\left(\frac{(K_{MJ}C_e)^{nMJ}}{1 + (K_{MJ}C_e)^{nMJ}}\right)^{M_{MJ}/n_{MJ}} \tag{46}$$

Where, n_{MJ} and M_{MJ} are parameters characterizing the adsorbent surface heterogeneity. If n_{MJ} and $M_{MJ} = 1$, this isotherm model reduces to Langmuir isotherm, while if $n_{MJ} = M_{MJ}$ then the isotherm reduces to Langmuir-Freundlich isotherm model. This isotherm model is recommended based on the supposition of the local Langmuir isotherm model and distribution of adsorption energies in the active sites on the adsorbent (Ramadoss and Subramaniam, 2018).

2.6. Five-parameter isotherm models

2.6.1. Fritz–Schlunder isotherm model

A five-parameter empirical model has been developed that can perform more precise simulation of the model variations for application over a wide range of equilibrium data (Ayawei et al., 2017). This isotherm model is expressed as follows:

$$q_e = \frac{1_m FS_s K_1 C_e^{\alpha FS}}{1 + K_2 C_e^{\beta FS}} \tag{47}$$

When α_{FS}, and $\beta_{FS} = 1$, this model approaches Langmuir isotherm model, while for higher adsorbate concentrations it reduces to Freundlich isotherm model (Hamdaoui and Naffrechoux, 2007; Vijayaraghavan, 2015).

Isotherm models, their linear and nonlinear equations as well as parameters to plot the equation are summarized in Table 2. The linear and non-linear mathematical expressions of various isotherm models are summarized in Table 2.

3. Multi-parametric adsorption isotherm models

Since there is more than one component in the adsorption process, there should be an equilibrium and interaction between the components. The previously discussed isotherm models were for the description of mono-parametric adsorption systems. Hence, studying multi-parametric systems requires new adsorption isotherm models to adequately represent the adsorption processes. Since multi-parametric models were extrapolated from the mono-parametric isotherm models, so their classification is based on the models they were extended from (Padilla-Ortega et al., 2013). Various multi-parametric adsorption isotherm models are illustrated in Table 3.

4. The solid-water distribution coefficient K_{id}

Assessment of the association between a compound and solid phase at equilibrium in a given system can be achieved by knowing the ratio of total equilibrium concentration of the compound in the aqueous solution and in solids, which is denoted as K_{id}. Commonly, the value of K_{id} is determined by measuring the sorption isotherms in batch experiments. Furthermore, in practical applications, the constancy of K_{id} over some concentration range is equal to presuming either: i) linear isotherm describes the overall process, ii) multiplying the relative concentration variation by (n_i-1) will give a small value of the relative K_{id} variation, confirming the small value of the relative concentration variation. The K_{id} can be expressed as follows,

$$K_{id} = \frac{C_{ioc}\,.\,f_{oc} + C_{imin}\,.\,A_{surf} + C_{iex}\,.\,\sigma_{surf\,ex}\,.\,A_{surf} + C_{irxn}\,.\,\sigma_{surf\,rxn}\,.\,A_{surf}}{C_{iw,neut} + C_{iw,ion}} \tag{42}$$

Furthermore, evaluating the ability of the natural organic materials for the sorption of organic pollutants starts with defining an organic carbon normalized sorption coefficient in case it is assumed that the organic matter is the dominant sorbent. The "organic carbon normalized" solid-water partition coefficient (K_{ioc}) is different for each chemical.

$$K_{ioc} = \frac{K_{id}}{F_{oc}} = \frac{C_{ioc}}{C_{iw}} \tag{43}$$

Temperature dependence of K_{ioc} over a narrow temperature range can be expressed by:

$$lnK_{ioc} = \frac{\Delta_{POM\,w}H_i}{R}.\ \frac{1}{T} + const. \tag{44}$$

In which:

$$\triangle_{POM\,w}H_i = H^E_{iPOM} - H^E_{iw} \tag{45}$$

Where, H^E_{iPOM} and H^E_{iw} are i excess enthalpies in sorbed and aqueous phase, respectively. *POM* is the sorption from water to solid-phase organic matter. Since H^E_{iPOM} depends on considered concentration range. H^E_{iPOM} can be different at low concentration where sorption to specific sites could dominate the overall K_{ioc} when compared to higher concentrations where the major process is partitioning. It is assumed that $\Delta_{POM\,w}H_i$ is primarily determined by H^E_{iw} in case of sorption of weakly polar and apolar compounds hydrophobic sites, which includes materials of carbonaceous surfaces. Moreover, H^E_{iPOM} could significantly change with the concentration of polar compounds, which might undergo H-bond interactions specific sites in *POM*. To sum up, partition coefficient is not affected by temperature in which it increases by a factor of two when the temperature decreases by 10 degrees. As a result, it is expected that the isotherm nonlinearity will increase with decreasing the temperature (Schwarzenbach et al., 2003a,b; Carballa et al., 2008; Ternes et al., 2004; Poerschman and Kopinke, 2001; Golet et al., 2003).

5. Thermodynamic properties and parameters

Thermodynamic properties play a crucial role in an adsorption process to determine the spontaneity of the process. Both entropy (ΔS°) and enthalpy (ΔH°) should be considered for the determination of ΔG°. If ΔG° is a negative value at a given temperature, the reaction occurs spontaneously. Adsorption is considered as exothermic reaction when ΔH° is a negative value, while it is an endothermic reaction if ΔH° is a positive value. The affinity of the adsorbent towards the adsorbate is reflected by the positive ΔS° value, suggesting the increased randomness (Saha and Chowdhury, 2011). Thermodynamic constants can be estimated from the Langmuir isotherm model by using the following equations: $\Delta G^0 = \Delta H^0 - T\Delta S^0$ and Eq. (18), where Eq. (18) can be used to directly calculate ΔG°, while from van' Hoff slop and intercept the values of ΔH° and ΔS° can be predict (Adelodun et al., 2016). It is suggested by the thermodynamic relation between ΔG°, ΔH° and ΔS° that either (i) ΔH° or ΔS° are negative and $\Delta H^\circ > T\Delta S^\circ$, (ii) ΔH° or ΔS° are positive and $T\Delta S^\circ > \Delta H^\circ$, (iii) ΔH° is negative and ΔS° is positive (Saha and Chowdhury, 2011; Ullah et al., 2020)

6. Applications of adsorption isotherms

Determination of the amount of adsorbed material of a given pollutant can be achieved by using the adsorption isotherms, which describes the relationship of the adsorbent-adsorbate equilibrium. Fig. 4 illustrates the best-fit model for the experimental data of different adsorbents and adsorbates.

Recently, serious pollution has been caused to the environment due to several different industrial activities. Pollutant management can be achieved by adsorption, which is known to be one of the most effective techniques with high flexibility. Activated carbon is the most common adsorbent, but its usage is limited because it has a high cost, as well as problems related to disposal and regeneration (Uddin, 2017). Therefore, agricultural wastes are a highly vulnerable adsorbent due to its porous and loose structure, as well as having various reactive functional groups including hydroxyl and carboxyl groups. The following Figs. (Fig. 5–8Figure 5) represents various agricultural wastes, which were used as an adsorbent by different studies for the removal of several different pollutants (Dai et al., 2018). Moreover, CO_2 fixed bed adsorption on activated carbon (AC) was investigated by Ibrahim and Al-Meshragi (2019), results showed that the maximum sorption capacity of CO_2 onto AC with initial concentrations 10 and 13.725 %vol. ranged from 109.5–35.46 and from 129.65 to 35.55 mg CO_2/ g AC, respectively. The Langmuir isotherm model was the best-fit model of the experimental data. Hauchhum and Mahanta (2014), studied the adsorption of CO_2 on various adsorbents including AC, zeolite 13X and zeolite 4A. The results concluded that the AC had the lowest capacity while zeolite had the highest adsorption capacity at temperatures ranging between 25 and 60 °C and up to 1 bar pressure. Experimental data of zeolite 13X and zeolite 4A were best fitted to the Langmuir isotherm model, while Freundlich model was the best-fit model of AC experimental data. Rong et al. (2017), utilized silica gel for the adsorption of toluene, CO_2 and CH_4 as gaseous air pollutants and found that the adsorption capacity of silica gel was much higher for toluene than CH_4 and CO_2 due to the different surface interactions. The Langmuir isotherm model was used for understanding the adsorption process and was the best-fit model for the experimental data. Pirngruber et al. (2013), utilized zeolite NaX for CO_2 capture through vacuum swing adsorption (VSA) process and results showed that CO_2 purity was higher than 95 % and the recovery was more than 90 %, but regeneration process requires very low pressure, which could be a handicap for applying it in larger scale. This study used the Langmuir model to fit the experimental data as the best-fit model. Shafeeyan et al. (2015), investigated CO_2 adsorption equilibria on AC and ammonia-

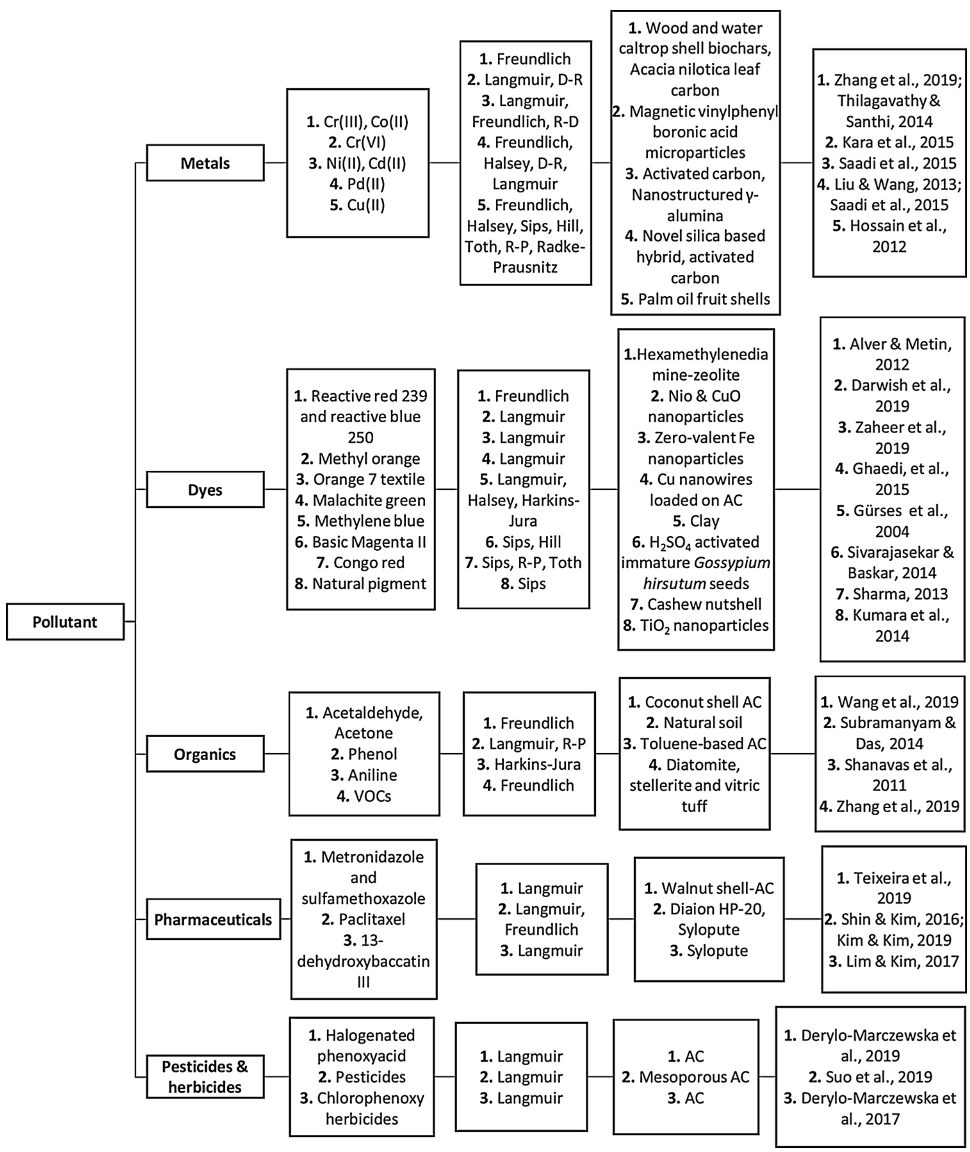


Fig. 4. The best-fit isotherm models for different adsorbents and adsorbates.

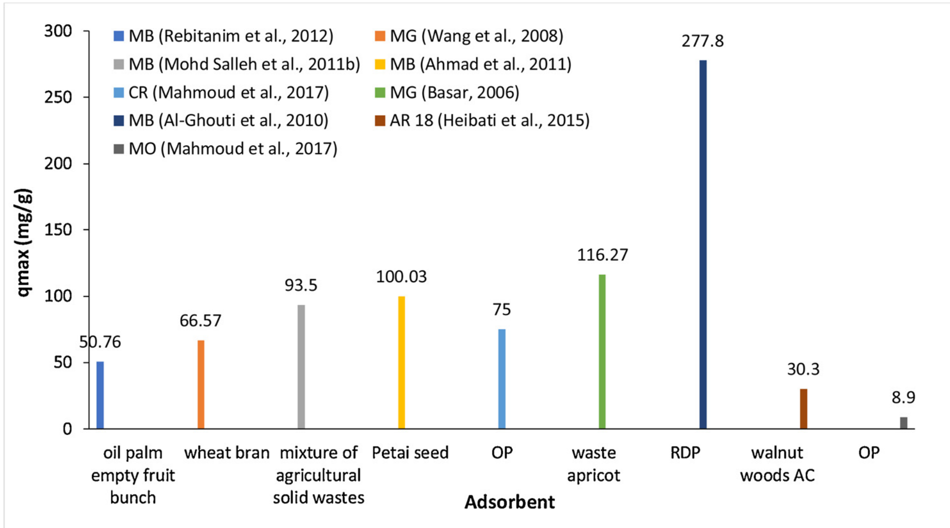


Fig. 5. Various agricultural waste adsorbent for the adsorption of different dyes.

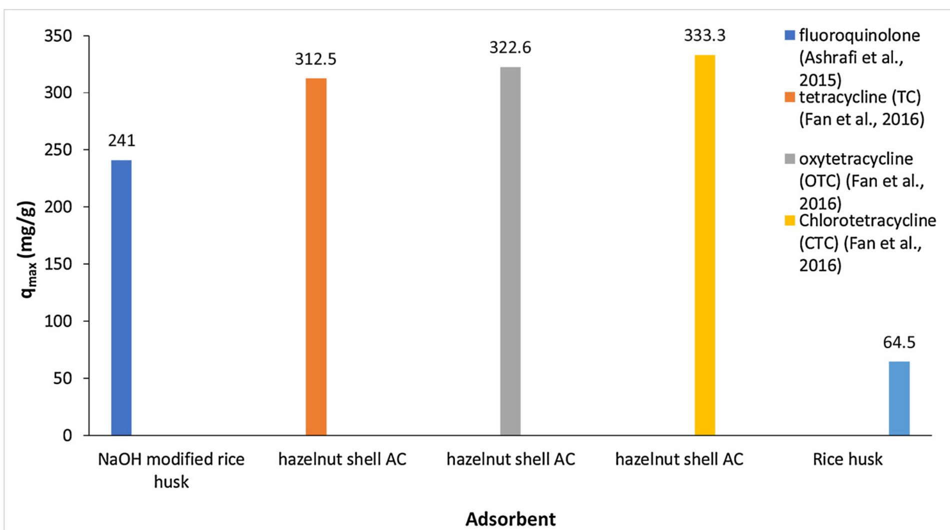


Fig. 6. Various agricultural waste adsorbent for the adsorption of different organic contaminants.

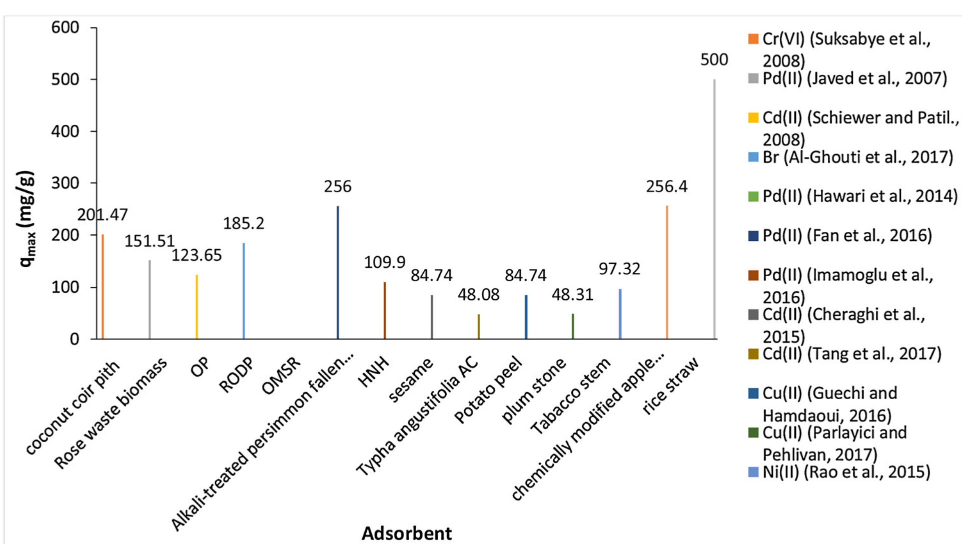


Fig. 7. Various agricultural waste adsorbent for the adsorption of different metals.

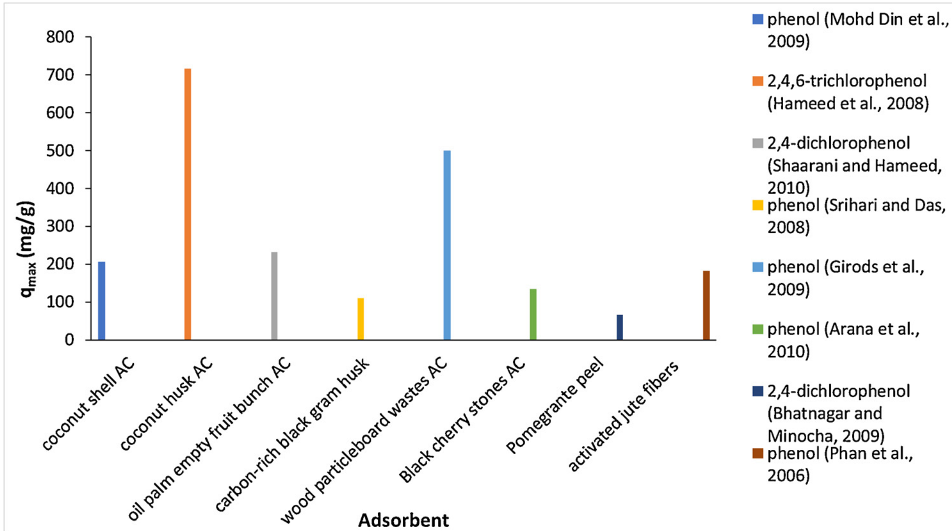


Fig. 8. Various agricultural waste adsorbent for the adsorption of different phenols.

modified AC and found that modified AC had higher CO_2 adsorption compared to AC at low partial pressure. Experimental data were fitted to various adsorption isotherm models and the Toth isotherm was the best-fit model of the obtained equilibrium data.

7. Applications of multi – parametric adsorption isotherm models

Nitzsche et al. (2019) investigated the lignin separation from hemicellulose sugar out of the beech wood hydrolysate by various adsorbents as well as determining the best single and multi-component isotherm model. Results showed that the EF multi-component parametric was the best-fit model. Monroy-Figueroa et al. (2014), applied various different multi-component isotherm models to investigate the best fit model for the adsorption of Cd(II) and Ni(II) by citric acid modified - *Byrsonima crossifolia* and found that (ES) multi-component model was the best fit model for the obtained experimental data compared to other studied models including EL and N-ML. This can be attributed to the fact that the ES model considers the competition between ions as well as having a heterogeneous surface, in which these factors are not considered in the other studied models. Martín-Lara et al. (2016), investigated the binary biosorption of Pd(II) and Cu(II) by peel of pine cone and studied EL and ES multi-parametric models and found that ES model was the best fit for the experimental data. Soetaredjo et al. (2013) used rice straw as an adsorbent for the removal of Cu(II) and Pd(II) from the synthetic binary component system and found that the ESL model gave the most reasonable fitted data with R^2 of 0.96. However, ESL isotherm did not give an efficient fit for the real effluent system having more than two ions (Fe, Ni, Pd, Cu, Hg, Zn, Cd, Cr, and Mn) and it did not give a proper representation of the real adsorption system due to considering the competition between two ions. Wynnyk et al. (2017), synthesized high purity zeolite 4A for the adsorption of the most common undesirable impurity in sour gas such as H_2S, CH_4, COS and CO_2 along with corresponding parameters for the modified Toth isotherm model. Results showed that zeolite 4A had lower adsorption for CH_4 compared to CO_2 and H_2S, while COS had strong adsorption with low capacity due to the adsorption change for different small sulfur species. In this study, the modified D-R model was compared to the modified Toth model in terms of SSE and results

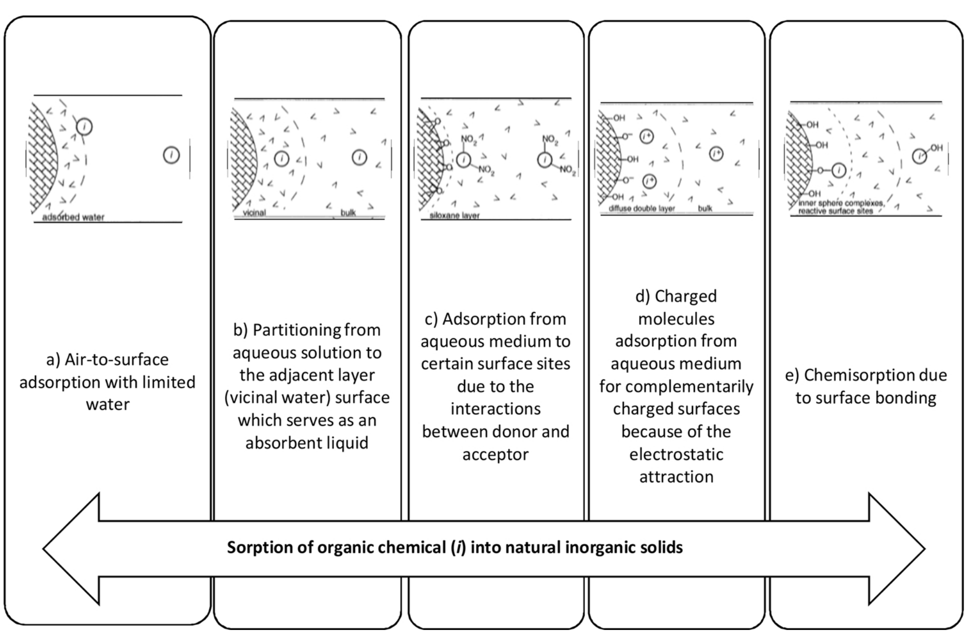


Fig. 9. Schematic views of different ways for sorption of organic chemical into natural inorganic solids (Modified from: Schwarzenbach et al., 2003a).

showed that the D-R model has limited utility due to the lack of temperature dependent terms. To sum up, it was found that zeolite 4A has the preferential to adsorb CO_2, COS, and H_2S up to their respective vapor pressures. In another study carried out by Wynnyk et al. (2019), adsorption of CH_4, CO_2, H_2S, and COS on mesoporous 60 Å silica gel. In this study, the adsorption data were fitted to the modified Toth equation that gave the lowest mean sum squared error (MSSE) compared to other semiempirical and empirical isotherms. Results revealed that an increased difference between the absolute and excess adsorption was noted due to the more intracrystalline volume of 60 Å silica gel. Compared to zeolite 4A, 60 Å silica gel has less capacity and affinity. Experimental study of air adsorption on gas-liquid interface in vapor condensation was performed by Zhang (2017), and results showed that the adsorption increases with increasing the relative air content and decrease with increasing the heat load, and it was found that air adsorption obeys Langmuir isotherm model. According to a study conducted by Dang et al. (2020), on the adsorption of methane on organic-rich shale, where the adsorption isotherms were measured at different temperatures by utilizing a volumetric experimental setup. Various excess models were used for the interpretation of the measured excess isotherms. It was found that the methane gas disorder at the gas-solid interface adsorption decreased due to the negative enthalpy value. DA + Henry model was the best excess adsorption isotherm.

8. Examples of adsorption mechanisms of different pollutants into various adsorbents

Fig. 9 explains the several methods for the sorption of organic chemical (*i*) to natural inorganic solids. Fig. 9a shows the strong attraction of organic sorbents of all polarities to dry inorganic surfaces from the air. Additionally, if the sorbates structure motivates the surface association, organics adsorption from aqueous medium to polar inorganic surfaces is of a great significance. As shown in Fig. 9b, that such sorption driving forces which includes the hydrophobicity of the sorbate discouraging the sorbate from staying in the bulk aqueous medium. Donor-acceptor interactions lead to rising specific sorbate-surface attractions, which is illustrated in Fig. 9c, while in Fig. 9d the presence of complementary charges on the two partners causes the specific sorbate-surface attraction to arise. These sorption mechanisms are considered as physisorption because there was no formation of covalent bond in the sorbate-surface associations. On the other hand, Fig. 9e illustrates the chemisorption process where the organic sorbate participate in the surface reactions of breaking and making the bonds. These interactions are described as inner sphere complexes due to the ability of organic sorbate and the surface to reach each other enabling the overlapping of the orbitals that are responsible for the bonding. This bonding process is significant when metals like iron or aluminum are exposed by the inorganic surface for the utilization of the organic sorbate as one of ligands (Schwarzenbach et al., 2003a,b; Bronner and Goss, 2011; Delle Site, 2001; Jagadamma et al., 2012).

According to Al-Ghouti et al. (2003), the proper understanding of the adsorption mechanism of basic and reactive dyes onto solid oxide surface has great importance for an effective dye removal from wastewater. The adsorption mechanism of dyes onto the diatomite surface containing silanol groups had been investigated as shown in Fig. 10. It is shown that there is an electrostatic attraction between reactive black (RB^-), reactive yellow (RY^-) and the positively charged diatomite surface at low pH value. On the other hand, at high pH, MB^+ has electrostatic (columbic) attraction with the negative charge on the diatomite surface. The adsorption mechanism of Br^- onto roasted date pits (RODPs) had been investigated by Al-Ghouti et al. (2017) and it is illustrated in Fig. 11. For metal ions, the determination of the complexes' stability is by the availability of the electron, i.e. the donor group basicity in which the greater the basicity, the greater the complex stability. According to Al-Ghouti et al. (2009), Cu^{2+} is able to form highly stable complexes regardless the donor group poor basicity. The two fixation types that would coexist are the cellulose-OH sites taking part in ion exchange, as well as those involved in adsorption and lignin-OH sites. Since fixation capacities depend on the adsorbed metal, Cu^{2+} has higher fixation capacities than Cd^{2+} due to the stronger affinity with hydroxyl functions onto the substrate. Hence, it is shown in Fig. 12 that A and D mechanisms are predominant for the adsorption of Cu^{2+} while B and C mechanisms are predominant for the adsorption of Cd^{2+}.

9. Criteria for choosing the optimum isotherm model and error analysis

The first criterion is that there should be a good fit between the isotherm function and the data. Determining the quality of the fit is achieved by the reduced chi-square statistics, which can be calculated by dividing the variance of the fit by the average variance of the data. Therefore, the reduced chi-square statistics should be less than the unity if the fitting function approximately equals the actual function. If the reduced chi-square statics has larger value, it indicates that there is poor fitting or there is greater data scattering than the expected, which is based on the experimental error bars (Parker and Garth, 1995). Secondly, the function of isotherm should be thermodynamically realistic. It should meet three necessities: i) when the concentration reaches zero, the isotherm should be linear; ii) at maximum concentration there should be finite capacity; and iii) for all concentrations, the slope should be positive. Henry's law represents the first requirement. Finite surface area and pore volume for adsorption are reflected by the second requirement. However, some isotherm models, e.g. Freundlich model, does not have an upper bound on the capacity. The importance of the third requirement to prevent the capacity decreasing as increasing the concentration, which is physically unrealistic, but it can happen with some polynomial isotherm models. Utility is the third criteria for the optimum isotherm model in which the isotherm should be usable for the various performed calculations, i.e., in ideal situations, the capacity can be analytically calculated from the concentration and vice versa. This is not always applicable, for example, concentration as a function of capacity is given by Myers isotherm model, but it is not possible to analytically invert the function. Therefore, a slow computational process that involves solving the equation's root can determine the capacity for a given concentration.

Recently, linear regression has been considered as one of the most viable ways to define the best-fitting relationship, to quantify the adsorbates' distribution, to verify the theoretical assumptions and the consistency of the isotherm models, as well as analysis of the adsorption system mathematically (Kumar, 2006; Lataye et al., 2008; Boulinguiez et al., 2008). Isotherm models fit better with the experimental data when the number of adsorption isotherm parameters increases in which the optimization of these parameters is accomplished by the

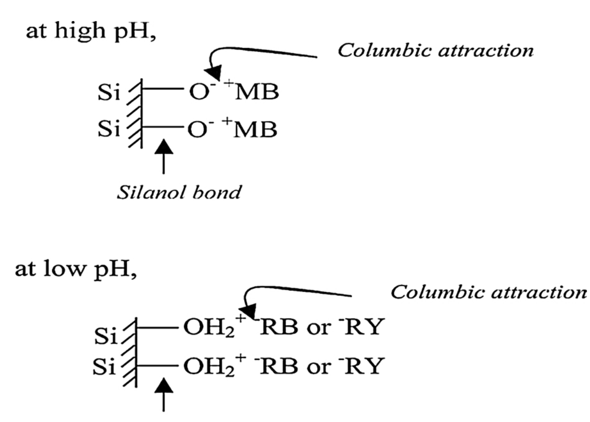


Fig. 10. Schematic representation of adsorption mechanism of MB, RB, and RY at high and low pH (Al-Ghouti et al., 2003).

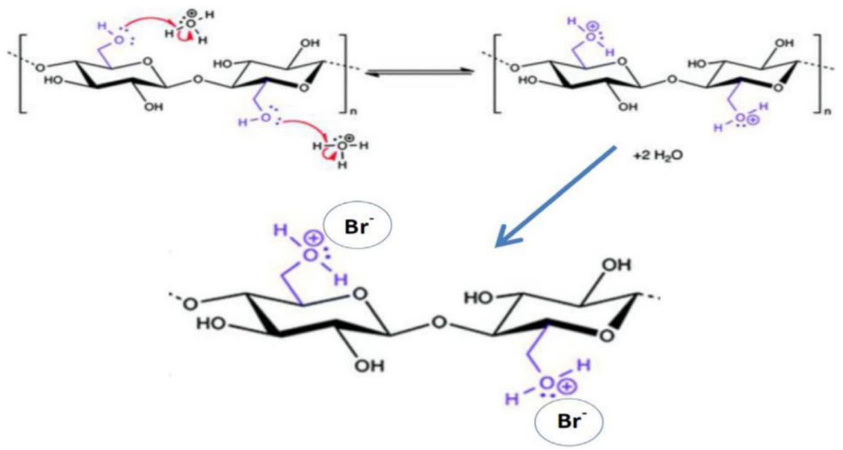


Fig. 11. The complexation mechanism of the outer-sphere surface of Br^- onto the cellulosic structure of RODPs.

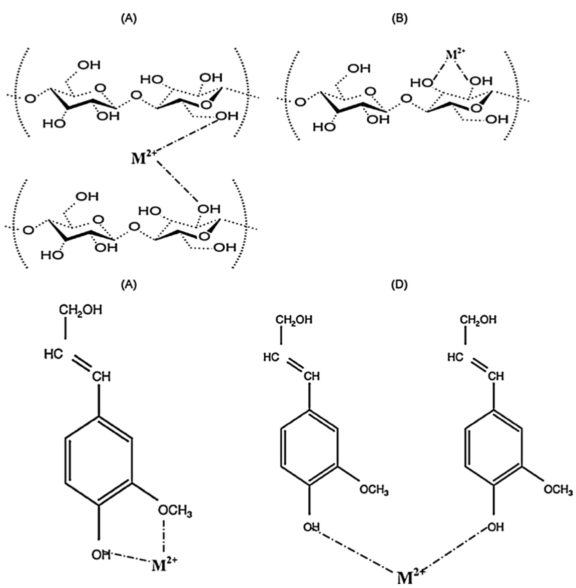


Fig. 12. The binding mechanism of RODPs, cellulose, lignin, respectively with Cu^{2+} and Cd^{2+}.

minimization process of the error function at given range of concentration. Thus, various mathematically error functions have been confronted drastically. This includes sum square error, sum of absolute errors, average relative error, Marquardt's percent standard deviation, Hybrid fractional error function, coefficient of determination, standard deviation of relative errors, nonlinear chi-square test, Spearman's correlation coefficient, coefficient of non-determination and sum of normalized errors which are discussed in Table 4 (Kumar et al., 2008). The error value will be a large number if the data from the isotherm model is different from the experimental data, and it will be small if they are similar. Minimizing the error values can be done by several methods such as the shuffled complex evolution (SCE) for the determination of the optimum parameter values for the isotherm models. Studies have shown that transformation of the adsorption isotherms to their linearized forms can cause a change in the error structure of the experimental data. Usually, nonlinear regression minimizes the distribution error between the predicted isotherm and the experimental data depending on its convergence criteria in which with the availability of the computer algorithms, this operation is not computationally difficult anymore (Ayawei et al., 2017).

Assuring that the model parameter values are optimized by performing a sensitivity analysis. Sensitivity analysis starts with (i) initialization of the model parameters with the minimum possible value of objective function i.e. SSE, (ii) nonlinear regression analysis is applied to find the optimal solution for minimizing the objective function, (iii) perturbation concept i.e. one parameter is modified as time and other parameters are kept constant, is used to implicate a sensitivity analysis to each model parameter, (v) re-estimation of the objective function is performed for each perturbation, (iv) a graph between the perturbation percentage of a respective model parameter and its objective function value is plotted. The global minimum is obtained if perturbation for all model parameters was 0%; otherwise, it indicates the poor nonlinear regression analysis of the values of the model parameters (Mukhtar et al., 2020).

10. Examples of linear and nonlinear forms of isotherm models

The number of independent parameters of an isotherm model determines its accuracy, and its mathematical simplicity is indicated by the relation of its popularity to the process application. Furthermore, linear regression analysis is widely useful in various adsorption data

Table 4
Error functions list.

Error function	Formula	Reference
Sum square error (ERRSQ)	$\sum_{i=1}^{n}(q_{e,calc}-q_{e,meas})_i^2$	(Foo and Hameed, 2010)
Sum of absolute errors (EABS)	$\sum_{i=1}^{n}\lvert q_{e,meas}-q_{e,calc}\rvert$	(Nebaghe et al., 2016)
Average relative error (ARE)	$\frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{q_{e,meas}-q_{e,calc}}{q_{e,meas}}\right\rvert$	(El Nemr et al., 2010)
Marquardt's percent standard deviation (MPSD)	$100\sqrt{\frac{1}{n-p}\sum_{i=1}^{n}\left(\frac{q_{e,meas}-q_{e,calc}}{q_{e,meas}}\right)^2}$	(Saadi et al., 2015)
Hybrid fractional error function (HYBRID)	$\frac{100}{n}\sum_{i=1}^{n}\left[\frac{q_{e,meas}-q_{e,calc}}{q_{e,meas}}\right]$	(Ayawei et al., 2017)
Coefficient of determination (R^2)	$r^2=\frac{(q_{e,meas}-\overline{q_{e,calc}})^2}{\sum(q_{e,meas}-q_{e,calc})^2+\overline{(q_{e,meas}-q_{e,calc})^2}}$	(Nebaghe et al., 2016)
Standard deviation of relative errors (S_{RE})	$\sqrt{\frac{\sum_{i=1}^{n}[(q_{e,meas}-q_{e,calc})_i-ARE]_i^2}{n-1}}$	(Saadi et al., 2015; Foo and Hameed, 2010)
Nonlinear chi-square test (X^2)	$\sum_{i=1}^{n}\frac{(q_{e,calc}-q_{e,meas})^2}{q_{e,meas}}$	(El Nemr et al., 2010)
Spearman's correlation coefficient (r_s)	$1-\frac{6\sum_{i=1}^{n}(q_{e,meas}-q_{e,calc})_i^2}{n(n-1)^2}$	(Ayawei et al., 2017)
Root mean square errors (RMS)	$100x\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(1-\frac{q_{e,calc}}{q_{e,isotherm}}\right)^2}$	(El Nemr et al., 2010)

and most frequently used to access the fits quality and the performance of the adsorption due to its simple equations. Recently, it has been noted that using the nonlinear optimization modeling has gained a development interest. Various researches investigated the possibility of applying nonlinear or linear isotherm models to describe the adsorption of heavy metals, dyes, and organic pollutants onto different low-cost adsorbents such as zeolites, bentonites, and activated carbon. Nouri et al. (2007), used wheat bran for the removal of cadmium ions from aqueous solution. Linear and nonlinear equations of the Langmuir isotherm model were utilized to determine the best-fit model and results showed that nonlinear method provided a better method to determine Langmuir parameters. Similarly, Kumar and Sivanesan (2007), compared the linear and nonlinear equations of Langmuir and Freundlich isotherm models for estimating the optimum parameters that best fit with the adsorption data of safranin onto rice husk at aqueous medium and results illustrated nonlinear method is more appropriate technique in predicting the optimum isotherm parameters compared to the linear method. Kundu and Gupta (2006), investigated the ability of iron oxide-coated cement for the sorption of arsenic. The best-fit isotherm model was determined by carrying out both linear and nonlinear regressions of Langmuir, Freundlich, Temkin, Dubinin-Radushkevich and Toth isotherm equations. Data evaluation was accomplished by using six error analysis methods including R^2, ERRSQ, EABS, ARE, HYBRID, and MPSD. Their values indicated the ability of Freundlich isotherm model to give the best-fit model for the overall experimental data under the studied concentration range. Ringot et al. (2007), investigated the in vitro adsorption of ochratoxin A (OA) onto the yeast biomass in which 7 adsorption models were examined to give the best toxin adsorption description. For linear models, R^2 magnitude was used to compare between the models, while SNE was utilized for both nonlinear and linear models. Based on the error function values, the isotherm constant values were consistent when they were measured by the linear and nonlinear calculations. Boulinguiez et al. (2008), used activated carbon for the adsorption of tetrahydrothiophene and used various error functions to determine the optimum parameters of the Langmuir isotherm model including R^2, S_{RE}, HYBRID, ARE, RSS, and EABS and found that nonlinear equation gives the best-fit for the experimental data. Kumar et al. (2008), compared both linear and nonlinear regression methods for the selection of the best-fit isotherm model of the experimental data for adsorption methylene blue by activated carbon. Six error functions (R^2, MPSD, HYBRID, ERRSQ, ARE, and EABS) were used to select the optimum nonlinear theoretical isotherm as well as determine the involved parameters in two and three parameter isotherms, while R^2 was used for the linear isotherm models. The results illustrated that MPSD is the best among other tested error functions regarding the minimization of the error distribution between the predicted isotherms and the experimental equilibrium data for two parameter isotherms, while R^2 was the best for three parameter isotherms. Moreover, it was found that while selecting the optimum isotherm, K^2 is very useful in determining the best error function. It was found that the nonlinear regression of Freundlich and Langmuir gave the optimum parameters. El-Khaiary (2008) studied the adsorption of methylene blue dye on water hyacinth and analyzed both the experimental data and the simulated adsorption equilibrium data by various least – squares regression including linear and nonlinear regression. The results showed that different adsorption isotherm parameters' estimates were produced by using different regression methods which in turn gives different conclusions about the adsorption mechanism and the adsorbent's surface properties. It was found that at low levels of random error, one linearized form of Langmuir isotherm and nonlinear regression of Langmuir gave accurate estimates of the isotherm parameters. Ho et al. (2002), investigated the sorption of copper, nickel and lead by peat from aqueous system, and analyzed the experimental data using various isotherm models including Langmuir, Freundlich, Temkin, Toth, and Sips isotherm models. Additionally, HYBRID, R^2, ERRSQ, MPSD, ARE, and EABS function errors were used to determine the best-fit isotherm model and error values illustrated that both linear and nonlinear regression of Sips equation gave the best model for the experimental data of the three divalent metal ions. Ayoob and Gupta (2008), examined the fluorine removal from aqueous solution by developing a novel adsorbent of hardened alumina cement granules (ALC) through isotherm fitting. It was found that with increasing the fluoride concentration, the adsorption capacity of ALC and the errors associated with the linearization in isotherm fitting also increases. It was suggested by X^2 analysis that there is a poor correlation between the equilibrium experimental data and the isotherm models at high concentration of fluoride for all tested models. Linear and nonlinear regression equations were used to determine the best-fit model and its optimum parameter through SNE values after optimizing the error functions. Based on the found results, from the linear and nonlinear regression procedure, Freundlich was the best-fit model. It was concluded from this study that instead of solely relying on R^2 values to determine the best-fit isotherm model, optimum isotherm parameters can be evolved by using the linear and nonlinear regression through the error functions. Ghodbane et al. (2008), investigated the removal of cadmium ions by eucalyptus bark from aqueous solution and used various isotherm models to describe the equilibrium data. The results showed that both the linearized and non-linearized forms of Freundlich isotherm model are the worst fit model. While the non-linearized form of the Langmuir isotherm model can be a better way than the linearized form of the Langmuir model to obtain the model parameters. After evaluating Pseudo-second-order model using both the linear and nonlinear curve fitting analysis method, results showed that the nonlinear method is the best to describe the sorption process. Han et al. (2009) applied various isotherm models to the experimental equilibrium data of methylene blue adsorption onto natural zeolites. Both linear and nonlinear regression methods were utilized to determine the relative parameters as well as using the error functions to determine which method was better in predicting the best-fit model for the experimental data. The results illustrated the suitability of both methods in obtaining the parameters with pseudo-second-order kinetic model being the best-fit model for the experimental data and in describing the adsorption mechanism. Mane et al. (2007), used bagasse fly ash for the adsorption of brilliant green dye. Freundlich, Langmuir, Redlich-Peterson, Temkin, and Dubinin-Radushkevich isotherm models were used for the analysis of the equilibrium isotherms for the adsorption process through both linear and nonlinear regression techniques. In this paper, various error functions were used for the determination of the best-fit model and its optimum parameters including ARE, HYBRID, MPSD, and ERRSQ. Results found that both linear and nonlinear forms of Langmuir and Redlich-Peterson isotherm models were the best in representing the experimental equilibrium data of the adsorption process. Ofomaja and Ho (2008), examined the adsorption of methyl violet dye from aqueous medium onto Mansonia wood sawdust and analyzed the equilibrium biosorption data by Langmuir, Freundlich and Redlich-Peterson isotherm models. The fits of three linear forms of the Langmuir isotherm model, the Freundlich model as well as Redlich-Peterson model were obtained through linear and nonlinear methods. Results showed that there was dissimilarity between Langmuir isotherm parameters from its three studied forms when the linear method was used, but not when the nonlinear method was utilized. The best fit of the experimental equilibrium data was obtained with both Redlich-Peterson and Langmuir isotherm models.

11. Conclusion

This review paper aimed to review various adsorption isotherm models including both mono-parametric and multi-parametric models with showing their applications as well as illustrating the best-fit model for various adsorbates and adsorbents. Moreover, the criteria for choosing the optimum isotherm model for the adsorption process with the error functions has been discussed. Based on our knowledge and findings the best fit models for: i) pesticides, herbicides, and pharmaceuticals are Langmuir and Freundlich isotherm models; ii) organics are Freundlich, Langmuir, Harkins-Jura and R-P models; iii) for dyes and metals are Freundlich, Langmuir, Halsey, Harkin-Jura, Sips, Hill, D-R, R-P, and Toth models. Before choosing the best optimum isotherm model for the equilibrium data there are certain criteria that should be considered, including, i) equilibrium data and the isotherm function should have good fit; ii) isotherm function should be thermodynamically realistic; and iii) in ideal situation, analytical calculation of capacity from concentration and vice versa, should be possible.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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